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# Influence of steady viscous forces on the non-linear behaviour of cantilevered circular cylindrical shells conveying fluid

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ABSTRACT

In this study, the effect of steady viscous forces (skin friction and pressurization) on the non-linear behaviour and stability of cantilevered shells conveying fluid is investigated for the first time. These forces are obtained by using the time-mean Navier–Stokes equations and are modelled as initial loadings on the shell, which are in a membrane-state of equilibrium with in-plane stresses. The unsteady fluid-dynamic forces, associated to shell motions, act as additional loadings on this pre-stressed configuration; they are modelled by means of potential flow theory and obtained by employing the Fourier transform technique. The problem is formulated using the extended Hamilton's principle in which the shell model is geometrically non-linear and based on Flügge's thin shell assumptions. This model includes non-linear terms of mid-surface stretching and the non-linear terms of curvature changes and twist, as well. The displacement components of the shell are expanded by using trigonometric functions for the circumferential direction and the cantilevered beam eigenfunctions for the longitudinal direction. Axisymmetric modes are successfully incorporated into the solution expansion based on a physical approximation. The system is discretized and the resulting coupled non-linear ODEs are integrated numerically, and bifurcation analyses are performed using the AUTO program. Results show that the steady viscous effects diminish the critical flow velocity of flutter and extend the range of flow velocity over which limit cycle responses are stable. On the other hand, the non-linear terms of curvature changes and twist have very little effect on the dynamics. The system exhibits rich post-critical dynamical behaviour and follows a quasiperiodic route to chaos.

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## 1. Introduction

Many biological and engineering systems involve thin-walled shells conveying incompressible fluid flow. Pulmonary passages and veins are examples of these types of structures in physiological systems; heat exchangers, jet pumps and heat shields of jet engines are examples of where such systems may be found in engineering applications.

Given the numerous applications of thin shells conveying fluid, studying their stability and acquiring knowledge about their post-instability behaviour are of great importance in avoiding catastrophic structural failures in engineering systems and in obtaining a better understanding of how biological systems function. A complete account and treatment of this subject may be found in Païdoussis [1, Chapter 7].

Païdoussis and Denise [2] developed the first linear analytical model for clamped–clamped and cantilevered (clamped-free) shells conveying inviscid incompressible fluid, and they also performed experiments. They showed for the first time that at sufficiently high flow velocities shells lose stability in either beam or shell modes: cantilevered shells lose stability by flutter, while supported-end shells do so by divergence (buckling). Since then, a large body of research has been devoted to the study of the linear stability of thin shells subjected to inviscid subsonic flows. Weaver and Unny [3] and Shayo and Ellen [4] investigated the stability of shells with simply-supported ends. Later, Shayo and Ellen [5] studied the linear dynamics of cantilevered shells conveying fluid, recognizing the importance of fluid behaviour beyond the free end of the shell; they introduced the concept of a "downstream flow model".

In all aforementioned studies the fluid viscosity is neglected, and fluid-dynamic forces are modelled assuming that the flow is inviscid. However, in reality fluids are viscous to a greater or lesser extent, and it is of interest to take the fluid viscosity into account and study its effects.

Païdoussis et al. [6] used the time-mean Navier–Stokes equations in conjunction with a linear shell model with clamped–clamped boundary conditions to study the primary aspects of the fluid viscosity, namely the steady-state pressurization (to overcome pressure drop) and the skin frictional force. These forces

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were found to be important as far as stability of the system is concerned. The effect of steady viscous forces on the linear stability of cantilevered shells subjected to either internal or annular flow was analysed by Païdoussis et al. [7]. They found that, in the case of internal flow, these forces have slight stabilizing effect, which becomes more pronounced for annular flow. Nguyen et al. [8] studied the effect of unsteady viscous forces on the linear stability of coaxial cantilevered shells conveying fluid in the annulus by means of a CFD-based model; they found that the unsteady effects of viscosity diminish with decreasing annular gap size, provided that the gap is small enough. Amabili and Garziera [9] studied the effect of steady viscous forces on the linear vibrations of simply-supported shells with non-uniform constraints, lying on an elastic foundation with added masses.

All the aforementioned theoretical work was done by means of linear theory. The linear theory, however, is limited to predicting the dynamics only up to the first loss of stability and is not capable of predicting the post-instability behaviour of the system. Non-linearities come into play at deformation amplitudes of the order of the shell thickness, which in many cases are not hard to achieve. Thus, investigation of the non-linear behaviour of the system is of value from a design point of view.

Investigation of the non-linear behaviour of simply-supported shells conveying inviscid fluid flow has been carried out by Lakis and Laveau [10]. They considered only the non-linearities associated with the fluid flow and found that these non-linearities do not have a considerable effect on the oscillations of the order of the shell thickness. Amabili et al. [11] re-examined the non-linear dynamics and stability of simply-supported shells conveying inviscid fluid. They used Donnell's non-linear shallow-shell theory for the structure and linearized potential flow theory for the fluid. It was shown that the system loses stability through a subcritical divergence. Also, the response of simply-supported shells containing quiescent or flowing fluid under harmonic excitation has been addressed in [12]. Non-linear dynamics and stability of clamped–clamped shells subject to annular or internal inviscid fluid flow was investigated theoretically and experimentally in [13] for the first time. They employed linearized potential flow theory to formulate the fluid-dynamic forces and Donnell's non-linear shallow-shell theory for the structure; comprehensive experiments were also performed. Both theory and experiments show that the system loses stability via a subcritical divergence. Amabili et al. [14] took into account the steady viscous effects in their non-linear model for supported-end shells conveying fluid.

Paak et al. [15] investigated the non-linear stability and the post-instability response of cantilevered shells conveying inviscid fluid for the first time. They found that the system loses stability by flutter in a supercritical fashion (*i.e.*, a supercritical Hopf bifurcation); multiple branches of stable limit cycles were found. The amplitudes of these limit cycles grow with flow velocity until they lose stability and the response becomes non-periodic (*e.g.*, quasiperiodic or chaotic).

One difficulty arising when dealing with shells conveying fluid is the inclusion of axisymmetric modes in the solution expansion. For supported-end shells, this issue is circumvented by assuming an infinite shell with periodic supports and presuming that the shell deformation over $[L, 2L]$ is the reflection of that over $[0, L]$, $L$ being the shell length [13]. Such an artifice is not possible for cantilevered shells conveying fluid. However, in this paper, the axisymmetric modes are included in the solution expansion by using a physically sound approximation.

The influence of steady viscous forces on the non-linear stability and the post-instability dynamics of cantilevered shells conveying fluid had not been explored up to now. In the present study, we extend the theory and investigate the effect of such forces on the non-linear behaviour of the system.

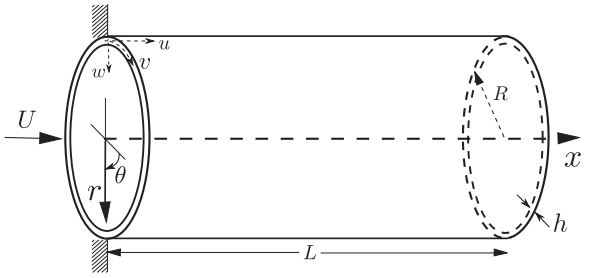



**Fig. 1.** Schematic of the system.

## 2. Formulation

### 2.1. Definitions and assumptions

Fig. 1 is a schematic diagram of the system under consideration. The shell has thickness $h$, length $L$, mean radius $R$ and is assumed to be thin (*i.e.*, $h/R \ll 1$), clamped at $x=0$ and free (unsupported) at $x=L$. The displacement components of the shell middle surface along the axial, circumferential and radial directions are denoted by $u$, $v$ and $w$, which are functions of time and the middle surface coordinates of the shell $(x,y)$ in which $y = R\theta$. The shell material is considered to be linearly elastic, homogeneous and isotropic, with Young's modulus $E$, Poisson ratio $\nu_s$ and density $\rho_s$. Viscous damping with coefficient $c$ is considered to model the structural energy dissipation. The shell non-linearity is of geometric type, which is described by large deformation theory.

The fluid is assumed to be incompressible with density $\rho_f$, flowing in the positive $x$-direction with a constant mean velocity $U$.

The unsteady fluid-dynamic forces due to the shell motions are derived by assuming that the fluid flow is inviscid and irrotational, thus enabling the utilization of potential flow theory. These forces are in the radial direction. The pressurization and the skin frictional forces (steady viscous forces) are taken into account as additional radial and axial loadings related to the steady mean flow. This is a simplification which renders the two parts (*i.e.*, the mean flow and perturbation flow fields) decoupled from the outset.

In this paper, partial derivatives may be represented by subscripts preceded by a comma, *e.g.*, $\partial^2()/\partial x \partial y \equiv ()_{,xy}$, and the prime denotes differentiation with respect to the argument of the primed function.

### 2.2. Equations of motion

The extended Hamilton's principle is utilized to obtain the governing equations of the system; *i.e.*,

$$\int_{t_1}^{t_2} [(\delta\mathcal{T} - \delta\mathcal{U} + \delta\mathcal{W}_d) + \delta\mathcal{W}_f]\, \mathrm{d}t = 0, \tag{1}$$

where $\delta$ is the variational operator, $\mathcal{T}$ the shell kinetic energy, $\mathcal{U}$ the elastic strain energy, $\delta\mathcal{W}_d$ the virtual work done by structural damping forces, and $\delta\mathcal{W}_f$ the virtual work done by the unsteady fluid-dynamic forces.

The non-linear shell model is developed based on Flügge's thin shell theory [16] and contains the non-linear terms due to mid-surface stretching as well as the non-linear terms of curvature changes and twist. To be able to perform the calculations in the undeformed reference configuration, Green's strain tensor and the second Piola–Kirchhoff stress tensor (a work-conjugate pair) are used.

The kinetic energy of the shell may be expressed as

$$\mathcal{T} = \frac{1}{2}\int_0^L \int_0^{2\pi R} \rho_s h(u_{,t}^2 + v_{,t}^2 + w_{,t}^2)\, \mathrm{d}y\, \mathrm{d}x, \tag{2}$$

and the strain energy may be obtained by integrating the strain energy density over the shell volume; this is treated after steady viscosity-related resultants are obtained.

The virtual work done by the damping force may be written as

$$\delta\mathcal{W}_d = \int_0^L \int_0^{2\pi R} [-ch(u_{,t}\delta u + v_{,t}\delta v + w_{,t}\delta w)]\, \mathrm{d}y\, \mathrm{d}x, \tag{3}$$

and the virtual work of the unsteady hydrodynamic forces is given by

$$\delta\mathcal{W}_f = -\int_0^L \int_0^{2\pi R} p|_{r=R}\delta w\, \mathrm{d}y\, \mathrm{d}x, \tag{4}$$

where $p$ is the perturbation pressure and will be formulated in Section 2.2.4.

### 2.2.1. Viscosity-related forces

In the work by Païdoussis et al. [6], the time-mean Navier–Stokes equations have been utilized to obtain the steady viscous forces. Here, we present only an outline of the procedure.

It is assumed that the fluid flow is incompressible and in the fully-developed turbulent flow regime. The time-mean Navier–Stokes equations for this system may be expressed as [17]

$$\begin{aligned}
&\frac{1}{\rho}\frac{\partial \overline{p}}{\partial r} = -\frac{1}{r}\frac{\partial(r\overline{u_r'^2})}{\partial r} + \frac{\overline{u_\theta'^2}}{r},\\
&0 = -\frac{\partial \overline{u_r' u_\theta'}}{\partial r} - 2\frac{\overline{u_r' u_\theta'}}{r},\\
&\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x} = -\frac{1}{r}\frac{\partial(r\overline{u_r' u_x'})}{\partial r} + \frac{\nu}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}U}{\mathrm{d}r}\right),
\end{aligned} \tag{5}$$

where $\nu$ is the fluid kinematic viscosity, $U$ is the mean flow velocity and $u_x'$, $u_r'$ and $u_\theta'$ are the turbulent fluctuating velocity components in the $x$, $r$ and $\theta$ directions, respectively. The overbar denotes the time-mean.

After lengthy mathematical manipulations, the mean pressure distribution inside the shell is obtained as

$$\overline{p}(x,r) = -2\frac{\rho}{R}U_\tau^2 x - \rho\overline{u_r'^2} + \rho\int_R^r \frac{\overline{u_\theta'^2} - \overline{u_r'^2}}{r}\,\mathrm{d}r + \overline{p}(0,R), \tag{6}$$

in which $U_\tau$ is the shear velocity defined by $U_\tau^2 = -\nu(\mathrm{d}U/\mathrm{d}r)|_{r=R}$. Now, the initial radial load (pressurization) and the initial axial load (shear at the wall) denoted by $P_r$ and $P_x$, respectively, may be written as

$$P_r(x,R) = -2\frac{\rho}{R}U_\tau^2 x + \overline{p}(0,R), \quad P_x = \rho U_\tau^2. \tag{7}$$

The fluid is discharged into the atmosphere; thus, from the first of Eq. (7) it is deduced that $\overline{p}(0,R) = (2\rho/R)U_\tau^2 L$. The shear velocity is related to the mean-flow velocity according to the formula $U_\tau^2 = fU^2/8$, where $f$ is the Darcy friction factor which may be obtained using Colebrook's implicit expression [18]

$$\frac{1}{\sqrt{f}} = -2\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right). \tag{8}$$

In this expression, $\varepsilon$ is the size of the shell surface roughness, $D$ is the shell diameter, and $\mathrm{Re} = UD/\nu$ is the Reynolds number. To avoid solving this implicit equation, it suffices to do just one iteration of (8) with an initial value for $f$ obtained by the following approximate explicit equation [19]:

$$\frac{1}{\sqrt{f}} = -1.8\log_{10}\left[\left(\frac{\varepsilon/D}{3.7}\right)^{1.11} + \frac{6.9}{\mathrm{Re}}\right]. \tag{9}$$

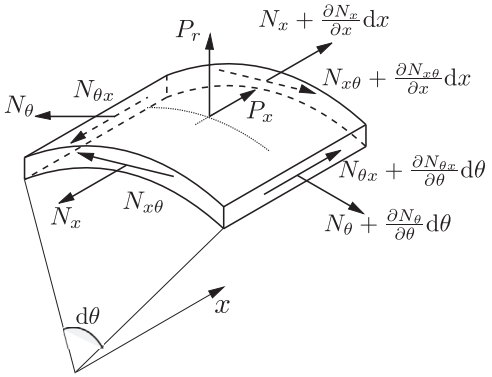



**Fig. 2.** An infinitesimal shell element in membrane-state of stress and under the action of external loads.

The accuracy of the friction factor, calculated by this scheme, is within $\pm 0.2\%$ of that of Eq. (8) alone, for $\varepsilon/D \le 0.05$ and $\mathrm{Re} \le 10^8$.

Finally, the initial loads may be expressed in the following concise functional form:

$$P_r = -(Cx + D), \quad P_x = B, \tag{10}$$

where $C = (f/4)\rho U^2/R$, $D = CL$ and $B = (f/8)\rho U^2$.

### 2.2.2. Initial stress resultants

Having obtained the initial loads acting on the system, we turn to finding the stresses in the shell under these loads. It is assumed that the shell is in a membrane-state of stress, and thus all the stress couples and transverse shear stress resultants are neglected. An infinitesimal element of the shell under external loads and internal stress resultants is shown in Fig. 2. In this figure, $N_\theta$, $N_x$, and $N_{x\theta}$ represent the hoop stress, axial stress and shear stress resultants, respectively.

Equations of equilibrium for this element in cylindrical coordinates $(r, y, x)$ in which $y = R\theta$ are found to be [6]

$$N_y = RP_r, \quad \frac{\partial N_{xy}}{\partial x} = -\frac{\partial N_y}{\partial y}, \quad \frac{\partial N_x}{\partial x} = -P_x - \frac{\partial N_{yx}}{\partial y}. \tag{11}$$

Since the system is axisymmetric, there is no dependence on the $y$ coordinate; hence, all derivatives with respect to $y$ vanish. Making use of (10), integrating the reduced system of PDEs and applying the boundary conditions $N_{xy}|_{x=0} = 0$ and $N_x|_{x=L} = 0$, the stress resultants are obtained as

$$N_x^I = -B(x - L), \quad N_y^I = -R(Cx + D), \quad N_{xy}^I = 0, \tag{12}$$

where the superscript $I$ indicates that they are associated with "initial stress".

### 2.2.3. Strain energy

The fluid-dynamic forces arising from the steady viscous effects are much larger than the unsteady inviscid forces. Thus, it is reasonable to model them as static loadings acting on the shell, leading to an initial internal stress field. The vibratory motion of the shell, then, occurs about this prestressed configuration and induces additional stresses on the structure. We assume that the initial stress field and the initial loads form a self-equilibrating system in a membrane-state of stress which is not affected by the vibratory stresses. Therefore, the initial stress field contributes to the equations of motion only through the non-linear part of the vibratory strains [20].

Taking the pre-stressed state of the shell as a reference level, variation of the strain energy of the shell may be written as

$$\delta\mathcal{U} = \int_V (\sigma_x\delta\varepsilon_x + \sigma_y\delta\varepsilon_y + \tau_{xy}\delta\gamma_{xy})\, \mathrm{d}V$$

$$+\int_V (\sigma_x^I \delta\varepsilon_x^{(2)} + \sigma_y^I \delta\varepsilon_y^{(2)} + \tau_{xy}^I \delta\gamma_{xy}^{(2)})\, \mathrm{d}V, \tag{13}$$

where the integration is performed over the shell volume $V$; $\varepsilon_x$, $\varepsilon_y$ and $\gamma_{xy}$ are the strain components and $\sigma_x$, $\sigma_y$ and $\tau_{xy}$ are the corresponding stress components, defined in Appendix A. The superscript "$I$" distinguishes the initial stresses from the additional or vibratory stresses.

The non-linear part of the strain components, designated by the superscript (2), is itself composed of two parts: (i) non-linear terms associated with mid-surface stretching ($\varepsilon_{x0}^{(2)}$, $\varepsilon_{y0}^{(2)}$ and $\gamma_{xy}^{(2)}$) and (ii) non-linear terms of curvature changes and twist, denoted by $\kappa_{()}^{(2)}$.

The first integral in (13) represents the variation of the strain energy of the shell under the action of unsteady fluid-dynamic forces, and the second integral is related to the steady fluid-dynamic forces via initial stresses.

Substituting Eqs. (2)–(4) and (13) into (1) and performing the necessary mathematical manipulations we arrive at

$$\int_0^L \int_0^{2\pi R} \{\mathcal{L}_1(\mathbf{u})\delta u + \mathcal{L}_2(\mathbf{u})\delta v + [\mathcal{L}_3(\mathbf{u}) + \mathcal{L}_{3f}(\mathbf{u})]\delta w\}\, \mathrm{d}y\, \mathrm{d}x - \int_0^{2\pi R} [\mathcal{R}_1(\mathbf{u})\delta u + \mathcal{R}_2(\mathbf{u})\delta v + \mathcal{R}_3(\mathbf{u})\delta w + \mathcal{R}_4(\mathbf{u})\delta w_{,x}]|_{x=L}\, \mathrm{d}y = 0, \tag{14}$$

where $\mathbf{u} = (u, v, w)$ is the displacement vector of the shell middle surface, $\mathcal{L}_{3f}(\mathbf{u}) = p(\mathbf{u}, r)|_{r=R}$ and

$$\mathcal{L}_i = L_i + L_i^C + L_i^I, \quad i = 1,2,3,$$
$$\mathcal{R}_i = R_i + R_i^C + R_i^I, \quad i = 1,2,3,4, \tag{15}$$

in which $L_i$ and $R_i$ are the differential expressions obtained if only the non-linear terms of mid-surface stretching are taken into account; these lengthy expressions are given in Paak et al. [15] and will not be repeated here. Differential expressions $L_i^C$ and $R_i^C$ are due to the non-linear terms of curvature changes and twist, and they are given in Appendix B. $L_i^I$ and $R_i^I$ are the contributions from initial stresses (or steady viscous forces), defined by

$$L_1^I = [N_x^I u_{,x}]_{,x} + [N_y^I u_{,y}]_{,y},$$
$$L_2^I = -\frac{1}{R}N_y^I\left(\frac{1}{R}v + w_{,y}\right) + [N_x^I v_{,x}]_{,x} + \left[N_y^I\left(v_{,y} - \frac{1}{R}w\right)\right]_{,y}, \tag{16}$$

$$L_3^I = -\frac{1}{R}N_y^I\left(\frac{1}{R}w - v_{,y}\right) + [N_x^I w_{,x}]_{,x} + \left[N_y^I\left(w_{,y} + \frac{1}{R}v\right)\right]_{,y},$$
$$R_1^I = N_x^I u_{,x}, \quad R_2^I = N_x^I v_{,x}, \quad R_3^I = N_x^I w_{,x}, \tag{17}$$

where the initial stress resultants $N_{()}^I$ are defined by (12). It should be noted that, since $N_x^I$ vanishes at $x=L$ (shell end), $R_i^I$ indeed does not contribute to the equations of motion.

Also, since the initial stresses are in a membrane state (*i.e.*, no bending moments and transverse shears), the non-linear terms of curvature changes and twist do not appear in $L_i^I$.

#### 2.2.4. Perturbation pressure

As discussed earlier, the perturbation pressure, due to shell motions, is obtained by assuming that fluid flow is inviscid and irrotational. It was shown by Evensen and Olson [21,22], Ginsberg [23], Gonçalves and Batista [24] and Lakis and Laveau [10] that at deformation amplitudes of the order of the shell thickness, a linear fluid model is adequate. Thus, linear potential flow theory is utilized to describe the flow field. The flow velocity $\mathbf{V}$ may be expressed as $\mathbf{V} = \nabla\Psi$, where $\Psi$ is a potential function defined by $\Psi = Ux + \Phi$; the first part is due to the mean undisturbed flow velocity $U$ in the $x$-direction, and the second term is the unsteady part, related to the shell motions. Continuity requires that the velocity field satisfies $\nabla \cdot \mathbf{V} = 0$; this results in the Laplace equation

$$\Phi_{,rr} + \frac{1}{r}\Phi_{,r} + \frac{1}{r^2}\Phi_{,\theta\theta} + \Phi_{,xx} = 0, \tag{18}$$

which is subject to the impermeability boundary condition at the shell–fluid interface,

$$\Phi_{,r}|_{r=R} = -(w_{,t} + Uw_{,x}). \tag{19}$$

The unsteady Bernoulli equation, $P/\rho_f + \frac{1}{2}|\mathbf{V}|^2 + \Psi_{,t} = P_s/\rho_f$, may be used to find the perturbation pressure. In this equation, $|\mathbf{V}|^2 = V_x^2 + V_r^2 + V_\theta^2$, $P$ is the fluid static pressure and $P_s$ is the stagnation pressure. Similarly to $\Psi$, $P$ may also be considered as being composed of a mean component $\overline{P}$ and the perturbation pressure $p$, *i.e.*, $P = \overline{P} + p$. Also, the stagnation pressure is given by $P_s = \overline{P} + \frac{1}{2}\rho_f U^2$. Substituting these ingredients back into the unsteady Bernoulli equation and neglecting second-order terms (*i.e.*, $|\mathbf{V}|^2 \simeq U^2 + 2U\Phi_{,x}$), the perturbation pressure is obtained as

$$p = -\rho_f(\Phi_{,t} + U\Phi_{,x}). \tag{20}$$

Hence, $p$ can be determined once $\Phi$ has been obtained.

## 3. Method of solution

The solution is expanded in the following form:

$$u = \sum_{n=0}^{\infty}\sum_{m=1}^{\infty} [A_{u,mn}(t)\cos(n\theta) + B_{u,mn}(t)\sin(n\theta)]\phi_m'(x),$$
$$v = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} [A_{v,mn}(t)\cos(n\theta) + B_{v,mn}(t)\sin(n\theta)]\phi_m(x),$$
$$w = \sum_{n=0}^{\infty}\sum_{m=1}^{\infty} [A_{w,mn}(t)\cos(n\theta) + B_{w,mn}(t)\sin(n\theta)]\phi_m(x), \tag{21}$$

where $A_{()}(t)$ and $B_{()}(t)$ are the time-dependent generalized coordinates, $n$ is the circumferential wavenumber, and $\phi_m(x)$ are admissible functions describing the shape of the shell deformation in the longitudinal direction. These functions must satisfy the geometric boundary conditions $u = v = w = 0$ and $w_{,x} = 0$ at $x=0$; in this study, cantilevered beam eigenfunctions are used for this purpose. The boundary conditions at $x=L$ ($R_i|_{x=L}$) are all natural and satisfied in the mean.

The solution expansion above comprises both asymmetric and axisymmetric modes ($n=0$). The latter takes effect at large amplitude vibrations of circular shells by providing a net inward contraction and thus avoiding the excessive in-plane stretching otherwise induced by other circumferential modes ($n \geq 2$) alone. The axisymmetric deformations of the shell in the circumferential direction are decoupled from the axial stretching and bending deformation of the shell and are therefore not taken into account in the modal expansion of $v$.

### 3.1. Solution for the perturbation pressure

In order to close the problem, the perturbation pressure needs to be expressed in terms of the shell displacements. For this purpose, the Fourier transform technique is employed; applying the Fourier transform to (18) results in

$$\Phi_{,rr}^* + \frac{1}{r}\Phi_{,r}^* + \frac{1}{r^2}\Phi_{,\theta\theta}^* - \alpha^2\Phi^* = 0, \tag{22}$$

where $\Phi^*$ is the Fourier transform of $\Phi$, given by

$$\Phi^*(\alpha, r, \theta, t) = \int_{-\infty}^{+\infty} \Phi(x, r, \theta, t)\mathrm{e}^{\mathrm{i}\alpha x}\, \mathrm{d}x. \tag{23}$$

Considering the finiteness of the solution at $r=0$, a solution to (22) is found to be

$$\Phi^*(\alpha, r, \theta, t) = \sum_{n=0}^{\infty} [c_{1n}(\alpha, t)\cos(n\theta) + c_{2n}(\alpha, t)\sin(n\theta)]\, \mathrm{I}_n(\alpha r), \tag{24}$$

in which $c_{1n}(\alpha,t)$ and $c_{2n}(\alpha,t)$ are yet to be determined by imposing the impermeability boundary conditions given in Eq. (19).

In the Fourier transform technique, the domain of the problem is $-\infty<x<+\infty$, which requires the flow conditions be specified over this range; it is reasonable to assume that there is no perturbations in the flow for $x<0$, whereas for $x>L$, on physical grounds, the perturbation pressure gradually decays to zero. This is accounted for by using an "outflow model" [5,7,25]. In this model the admissible functions are extended to $x>L$ in a way that the flow perturbations tend to zero smoothly over a distance beyond the free end of the shell. For that purpose, $w(x,\theta,t)$ is rewritten as

$$w(x,\theta,t)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}[A_{w,mn}(t)\cos(n\theta)+B_{w,mn}(t)\sin(n\theta)]f_m(x), \tag{25}$$

where $f_m(x)$ are the extended admissible functions given by

$$f_m(x)=\begin{cases}\phi_m(x) & \text{if } 0\le x\le L\\ R_m(x) & \text{if } L<x;\end{cases} \tag{26}$$

$R_m(x)$ represents the outflow model and has the functional form $R_m(x)=(Ax/L+B)e^{1-x/L}$, where $A=[\phi_m(L)+L\phi_m'(L)]$ and $B=[-L\phi_m'(L)]$; these are determined so that $f_m(x)$ be $C^1$-continuous at $x=L$. Enforcing the Fourier-transformed form of the impermeability condition (19), the solution is obtained as

$$\begin{aligned}\Phi^*(\alpha,r,\theta,t)=-\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}\Bigg\{&\left[\frac{\partial}{\partial t}A_{w,mn}(t)-i\alpha UA_{w,mn}(t)\right]\cos(n\theta)\\&+\left[\frac{\partial}{\partial t}B_{w,mn}(t)-i\alpha UB_{w,mn}(t)\right]\sin(n\theta)\Bigg\}\\&\times\frac{I_n(\alpha r)}{\alpha I_n'(\alpha r)}[\phi_m^*(\alpha)+R_m^*(\alpha)],\end{aligned} \tag{27}$$

where $\phi_m^*(\alpha)$ and $R_m^*(\alpha)$ are the Fourier transform of $\phi_m(x)$ and $R_m(x)$.

Applying the Fourier transform to (20) and making use of (27) yields

$$\begin{aligned}p^*(\alpha,r,\theta,t)&\\=\rho_f\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}\Bigg\{&\left[\frac{\partial^2}{\partial t^2}A_{w,mn}(t)-2i\alpha U\frac{\partial}{\partial t}A_{w,mn}(t)-\alpha^2U^2A_{w,mn}(t)\right]\cos(n\theta)\\&+\left[\frac{\partial^2}{\partial t^2}B_{w,mn}(t)-2i\alpha U\frac{\partial}{\partial t}B_{w,mn}(t)-\alpha^2U^2B_{w,mn}(t)\right]\sin(n\theta)\Bigg\}\\&\times\frac{I_n(\alpha r)}{\alpha I_n'(\alpha r)}[\phi_m^*(\alpha)+R_m^*(\alpha)].\end{aligned} \tag{28}$$

Taking the inverse Fourier transform of $p^*$ (*i.e.*, $(1/2\pi)\int_{-\infty}^{\infty}p^*(\alpha,r,\theta,t)e^{-i\alpha x}\,d\alpha$) and substituting the result into (4), the virtual work of unsteady fluid-dynamic forces may be expressed as

$$\delta\mathcal{W}_f=-\frac{1}{2\pi}\int_{-\infty}^{\infty}\int_0^L\int_0^{2\pi}p^*|_{r=R}\,\delta w\,e^{-i\alpha x}R\,d\theta\,dx\,d\alpha, \tag{29}$$

where

$$\delta w=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}[\delta A_{w,mn}(t)\cos(n\theta)+\delta B_{w,mn}(t)\sin(n\theta)]\phi_m(x). \tag{30}$$

Substituting (28) and (30) into (29) and after lengthy manipulations, (29) may be written as

$$\begin{aligned}\delta\mathcal{W}_f=\rho_fR\sum_{j=1}^{\infty}\sum_{m=1}^{\infty}&\left(-I_{jm0}^{(1)}\frac{\partial^2}{\partial t^2}A_{w,m0}+I_{jm0}^{(2)}U\frac{\partial}{\partial t}A_{w,m0}+I_{jm0}^{(3)}U^2A_{w,m0}\right)\\&\times\delta A_{w,j0}+\frac{\rho_fR}{2}\sum_{n=1}^{\infty}\sum_{j=1}^{\infty}\sum_{m=1}^{\infty}\Bigg[\Bigg(-I_{jmn}^{(1)}\frac{\partial^2}{\partial t^2}A_{w,mn}\\&+I_{jmn}^{(2)}U\frac{\partial}{\partial t}A_{w,mn}+I_{jmn}^{(3)}U^2A_{w,mn}\Bigg)\delta A_{w,jn}\\&+\left(-I_{jmn}^{(1)}\frac{\partial^2}{\partial t^2}B_{w,mn}+I_{jmn}^{(2)}U\frac{\partial}{\partial t}B_{w,mn}+I_{jmn}^{(3)}U^2B_{w,mn}\right)\delta B_{w,jn}\Bigg],\end{aligned} \tag{31}$$

in which $I_{jmn}^{(\cdot)}$ and $I_{jm0}^{(\cdot)}$ are integrals appearing due to the inverse Fourier transform and integration over the shell length. These integrals in dimensionless form are presented in Appendix C. In Eq. (31), the expression with a double summation is the contribution from axisymmetric modes ($n=0$), and the expression with a triple summation is due to asymmetric modes. The integrals $I_{jm0}^{(\cdot)}$, associated with axisymmetric modes, need to be treated in a special way, particularly $I_{jm0}^{(1)}$ which is related to the fluid added mass for $n=0$.

### 3.1.1. Integrals related to axisymmetric modes

Since the change of volume due to axisymmetric modes is one order of magnitude larger than that of asymmetric modes and the fluid domain is $-\infty<x<+\infty$, the integrals $I_{jm0}^{(1)}$, related to the fluid added mass, become unbounded. Hence, axisymmetric shell deformations lead to infinite kinetic energy of the fluid and thus infinite added mass. Mathematically, this occurs because the integrand of $I_{jmn}^{(1)}$ has an irremovable singularity at $\alpha=0$ when $n=0$. Therefore, for modes with $n=0$ the value of integrals $I_{jm0}^{(1)}$ must be obtained differently. To this end, we assume that when $n=0$ the fluid domain is $0\le x\le L$; in other words the fluid is contained by the shell. In this way, the integrals $I_{jm0}^{(1)}$ are approximated by finding the fluid-dynamic loadings of quiescent fluid inside the shell. Physically, this is a sound approximation since the fluid added mass does not depend on the flow velocity. The integrals $I_{jm0}^{(2)}$ and $I_{jm0}^{(3)}$ are obtained by using the expressions for $I_{jmn}^{(2)}$ and $I_{jmn}^{(3)}$, but setting the outflow length to zero, meaning that all flow perturbations for these components of the solution vanish when the fluid exits the shell at $x=L$.

Based on the forgoing assumption, the added mass associated with axisymmetric modes may be obtained by solving the Laplace equation, $\nabla^2\phi=0$, over $0\le x\le L$ and subject to the following boundary conditions:

$$\phi|_{x=0}=\phi|_{x=L}=0, \tag{32}$$

$$\phi_{,r}|_{r=R}=-w_{,t}. \tag{33}$$

The boundary condition in (32) simply states that there is no perturbation in the velocity potential at the shell ends, and (33) is the impermeability condition at the fluid–structure interface. Using the method of separation of variables a solution for the $n=0$ case, satisfying (32), is found as

$$\phi(x,r,t)=\sum_i^{\infty}c_{i0}(t)I_0\left(\frac{i\pi}{L}r\right)\sin\left(\frac{i\pi}{L}x\right), \tag{34}$$

where $c_{i0}(t)$ is an unknown function of time. Enforcing the impermeability condition (33) by using the expansion for $w$ defined in (21) and the solution (34), we find that

$$c_{i0}(t)=\frac{-2}{i\pi I_0'(i\pi R/L)}\sum_{m=1}^{\infty}\left(\int_0^L\phi_m(x)\sin\left(\frac{i\pi}{L}x\right)dx\right)\dot{A}_{w,m0} \tag{35}$$

The perturbation pressure may be obtained from (20) by setting $U=0$;

$$p(x,r,t)=\sum_{m=1}^{\infty}\sum_{i=1}^{\infty}F_{mi}\frac{2\rho_fI_0(i\pi\,r/L)}{i\pi I_0'(i\pi R/L)}\sin(i\pi x/L)\ddot{A}_{w,m0}, \tag{36}$$

where $F_{mi}=\int_0^L\phi_m(x)\sin(i\pi x/L)\,dx$. Making use of (4) and (36), the virtual work done via the fluid perturbation pressure may be

written as

$$\delta\mathcal{W}_f = \sum_{j=1}^{\infty}\sum_{m=1}^{\infty}\sum_{i=1}^{\infty} F_{mi}F_{ji}\frac{-4R\rho_f \mathrm{I}_0(i\pi R/L)}{i\mathrm{I}_0'(i\pi R/L)}\ddot{A}_{w,m0}\delta A_{w,j0}. \tag{37}$$

Comparing the above expression for $\delta\mathcal{W}_f$ with that given in (31), we understand that $I^{(1)}_{jm0}$ may be approximately expressed as

$$I^{(1)}_{jm0} = \sum_{i=1}^{\infty} F_{mi}F_{ji}\frac{4\mathrm{I}_0(i\pi R/L)}{i\mathrm{I}_0'(i\pi R/L)}. \tag{38}$$

It should not escape the reader's attention that only the $n=0$ case has been dealt with in an approximate way; the integrals related to asymmetric modes (*i.e.*, $n\neq 0$) are calculated without any approximations involved.

### 3.2. Non-dimensionalization and solution to the equations of motion

To render the equations of motion dimensionless, a reference time $T$ and a reference velocity $\mathcal{U}$ are defined as follows:

$$T = R\left[\frac{(1-\nu_s^2)\rho_s}{E}\right]^{1/2}, \quad \mathcal{U} = \left[\frac{E}{\rho_s(1-\nu_s^2)}\right]^{1/2}; \tag{39}$$

also, the following dimensionless parameters are introduced:

$$\overline{\alpha} = \alpha L, \quad \xi = \frac{x}{L}, \quad \theta = \frac{y}{R}, \quad \overline{t} = \frac{t}{T}, \quad \overline{u} = \frac{u}{R}, \quad \overline{v} = \frac{v}{R}, \quad \overline{w} = \frac{w}{R},$$
$$\overline{A}_{(),mn} = \frac{A_{(),mn}}{R}, \quad \overline{B}_{(),mn} = \frac{B_{(),mn}}{R}, \quad \overline{U} = \frac{U}{\mathcal{U}}, \quad \beta = \frac{R}{L}, \quad \varepsilon = \frac{h}{R},$$
$$k = \frac{1}{12}\varepsilon^2, \quad \gamma = \frac{\rho_f}{\rho_s}. \tag{40}$$

Using these dimensionless parameters, and henceforth dropping the over-bars for simplicity, the dimensionless form of (14) may be expressed as

$$\int_0^1\int_0^{2\pi}[\mathcal{L}_1(\mathbf{u})\delta u + \mathcal{L}_2(\mathbf{u})\delta v + \mathcal{L}_3(\mathbf{u})\delta w]\,\mathrm{d}\theta\,\mathrm{d}\xi$$
$$-\int_0^{2\pi}\{\beta[\mathcal{R}_1(\mathbf{u})\delta u + \mathcal{R}_2(\mathbf{u})\delta v + \mathcal{R}_3(\mathbf{u})\delta w] + \beta^2[\mathcal{R}_4(\mathbf{u})\delta w_{,\xi}]\}|_{\xi=1}\,\mathrm{d}\theta$$
$$+\delta\mathcal{W}_f = 0. \tag{41}$$

As mentioned earlier, the differential operators $\mathcal{L}_i$ are long-winded and are not repeated here for the sake of brevity; these operators in dimensionless form are defined in Paak et al. [15]. The dimensionless form of $\delta\mathcal{W}_f$ and the integrals involved therein are given in Appendix C.

The initial stresses in (12) in dimensionless form are

$$N_x^I = -\frac{\gamma}{\varepsilon\beta}\frac{f(U)}{8}U^2(\xi-1), \quad N_y^I = -\frac{\gamma}{\varepsilon\beta}\frac{f(U)}{4}U^2(\xi-1). \tag{42}$$

Making the necessary substitutions and performing the integrations in (41), the dimensionless equations of motion in discretized form may be written as

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} + \mathbf{f}(\mathbf{q}) + \mathbf{g}(\mathbf{q}) = \mathbf{0}, \tag{43}$$

where $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are the linear mass, damping and stiffness matrices, respectively; $\mathbf{q}$ is the vector of generalized coordinates, given by

$$\mathbf{q} = \{A_{u,m0}, A_{w,m0}, A_{u,mn}, B_{u,mn}, A_{v,mn}, B_{v,mn}, A_{w,mn}, B_{w,mn}\},$$
$$m = 1,\ldots,M, \quad n = 1,\ldots,N, \tag{44}$$

in which $M$ and $N$ are the number of axial and circumferential modes taken for this analysis. Furthermore, the vector functions $\mathbf{f}$ and $\mathbf{g}$ contain the non-linear terms of quadratic and cubic order, respectively.

The system of second-order non-linear ODEs in (43) is then cast into state-space form (*i.e.*, a system of first-order ODEs) and implemented as a Fortran function to be solved numerically.

Bifurcation diagrams of the shell deformation components versus dimensionless flow velocity are calculated using the AUTO 07p software. Time responses of the system and Poincaré maps are obtained using a Fortran program based on the Runge–Kutta 4th-order method and the Hénon algorithm [26]. In these programs, the Darcy friction factor is computed in a separate Fortran subroutine and fed into the main code. Fig. 3 shows the friction

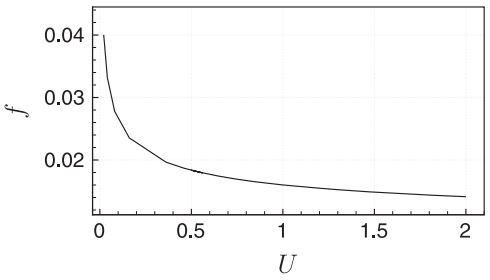



**Fig. 3.** Darcy friction factor versus dimensionless flow velocity.

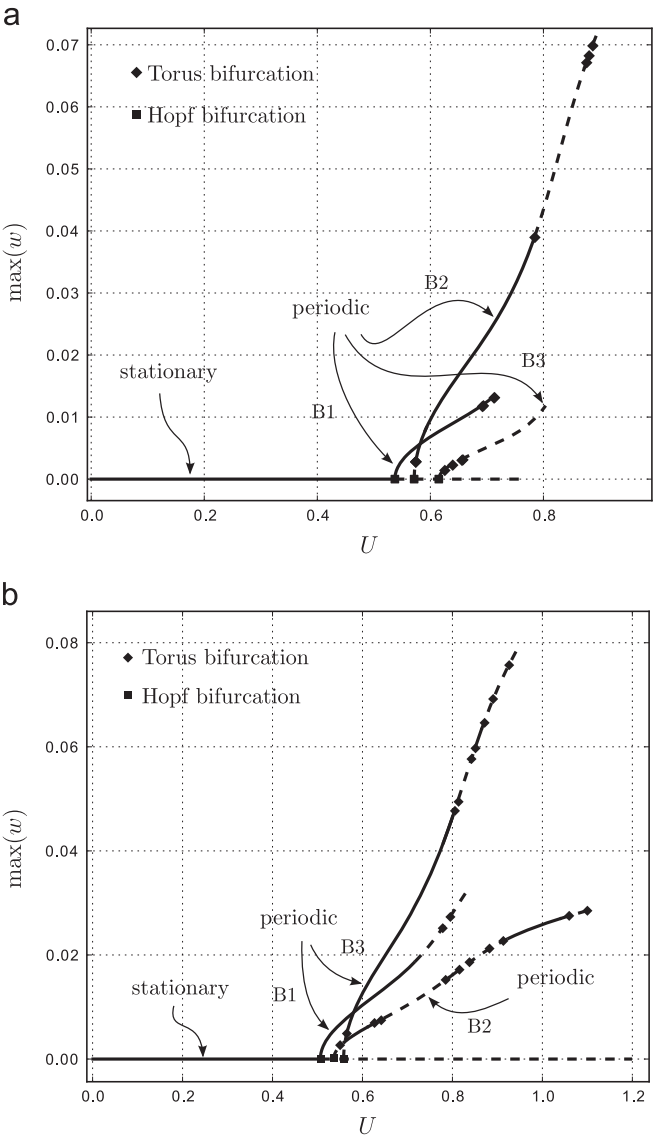



**Fig. 4.** Maximum value of the dimensionless radial deformation $w$ at $(\xi=1, \theta=\pi/3)$ versus dimensionless flow velocity $U$; ——, stable solution; – – – – – –, unstable solution; B1, branch 1, and so on; $L/R=3$, $h/R=0.02$; $n=0,4,5,6$ and $m=1,2$; (a) inviscid flow; (b) with steady viscous effect.

factor as a function of dimensionless flow velocity. Several diagnostics tools such as FFT analysis, phase-plane plot, probability density function, auto-correlation and Lyapunov exponents are utilized to characterize the non-linear behaviour of the system. Since the numerical solution of the system is computationally intensive, programs were run on a cluster in parallel or batch mode.

In the results that follow, all quantities are presented in dimensionless form.

## 4. Results and discussion

### 4.1. Non-linear terms of curvature changes and twist

Taking into account these non-linear terms increases the size of the already-large equations of motion substantially, resulting in extra computational cost. However, our analysis showed that the influence of the non-linear terms of curvature changes and twist on the behaviour of the system is in fact negligible; the results obtained are almost identical to what was obtained when considering only the non-linear terms of mid-surface stretching. Hence results with these extra non-linear terms are not presented here for the sake of brevity.

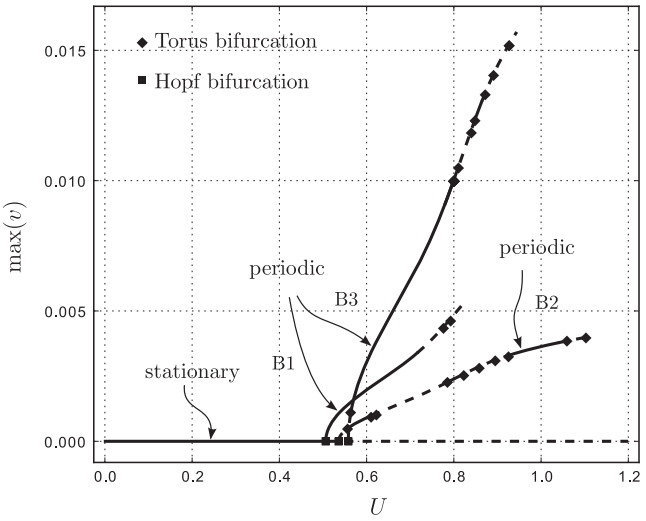



**Fig. 5.** Maximum value of the dimensionless circumferential deformation $v$ at $(\xi=1,\theta=\pi/3)$ versus dimensionless flow velocity $U$ for the system considering steady viscous effects; ——, stable solution; – – – – –, unstable solution; B1, branch 1, and so on; $L/R=3$, $h/R=0.02$; $n=0,4,5,6$ and $m=1,2$.

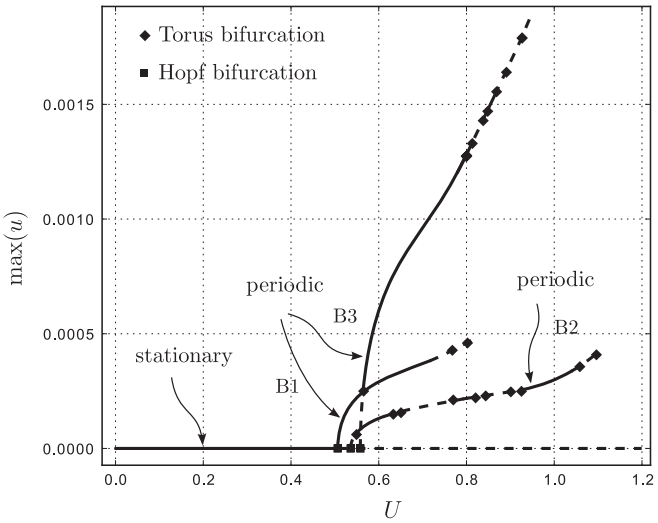



**Fig. 6.** Maximum value of the dimensionless axial deformation $u$ at $(\xi=1,\theta=\pi/3)$ versus dimensionless flow velocity $U$ for the system considering steady viscous effects; ——, stable solution; – – – – –, unstable solution; B1, branch 1, and so on; $L/R=3$, $h/R=0.02$; $n=0,4,5,6$ and $m=1,2$.

### 4.2. Steady viscous effect

To study the effect of the steady viscous forces on the non-linear dynamics of the system, we consider two sets of geometric parameters, namely $L/R=3$ and $L/R=7$, and in both cases $h/R=0.02$. In the calculations, the fluid-to-structure density ratio is taken to be $\gamma=0.00136$, the Poisson ratio is $\nu_s=0.5$, and the dimensionless viscous damping is set to $c=0.0001$. These parameter values match those used by Paak et al. [15], in the inviscid version of their theory. Additionally, for calculating the friction factor $f$, we assume that $\nu=1.511\times10^{-5}$ m$^2$/s and $\varepsilon=10^{-6}$ m.

For each geometry, in addition to the axisymmetric modes, the solution expansion in the circumferential direction is composed of the critical mode associated with the lowest critical flow velocity, predicted by linear theory, and the two adjacent modes; given the

**Table 1**
Dimensionless flow velocities associated with Hopf bifurcations of inviscid and viscous theory, $L/R=3$.

| Flow | HB1 ($U_{cr}$) | HB2 | HB3 |
|---|---|---|---|
| Inviscid | $U_{inv}=0.537$ | 0.571 | 0.614 |
| Viscous | $U_{vis}=0.506$ | 0.535 | 0.558 |
| $(U_{vis}-U_{inv})/U_{inv}$ | $\sim -6\%$ | $\sim -6\%$ | $\sim -9\%$ |

**Table 2**
The interval of $U$ for stable periodic orbits on branches B1, B2 and B3, predicted by inviscid and viscous theories (cf. Fig. 4), $L/R=3$.

| Flow | Branch B1 | Branch B2/B3 |
|---|---|---|
| Inviscid | $I_{inv}=(0.537,0.692)$ | (0.574, 0.784) |
| Viscous | $I_{vis}=(0.506,0.715)$ | (0.564, 0.8) |
| $(\lvert I_{vis}\rvert-\lvert I_{inv}\rvert)/\lvert I_{inv}\rvert$ | $\sim 35\%$ | $\sim 12\%$ |

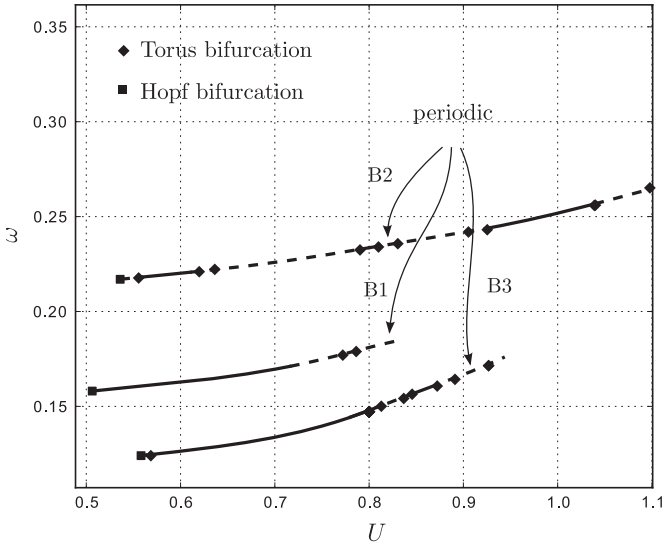



**Fig. 7.** Dimensionless frequency of limit cycles versus dimensionless flow velocity $U$ for the system considering steady viscous effects; ——, stable solution; – – – – –, unstable solution; B1, branch 1, and so on; $L/R=3$, $h/R=0.02$; $n=0,4,5,6$ and $m=1,2$.

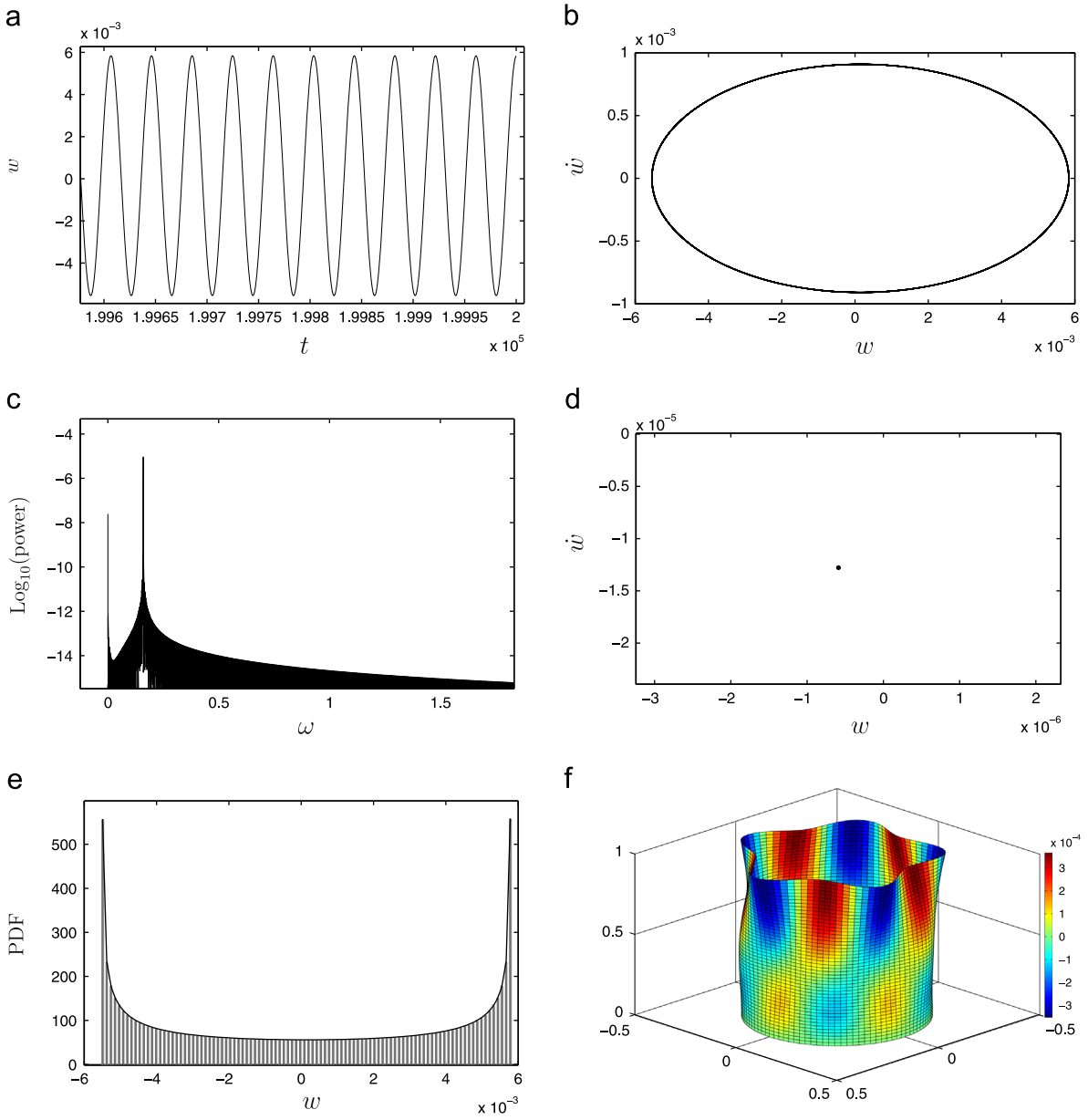



**Fig. 8.** Response of the system considering steady viscous effects at $U=0.54$ for $L/R=3$, $h/R=0.02$, $n=0,4,5,6$ and $m=1,2$: (a) time trace of $w$ ($\xi=1,\theta=\pi/3$); (b) phase portrait; (c) power spectrum; (d) one-sided Poincaré map; (e) PDF; (f) shell deformation shape.

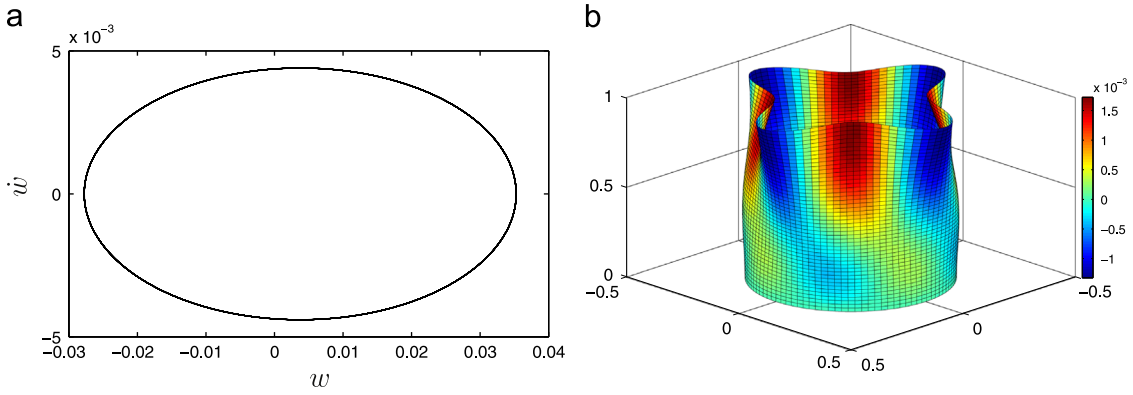



**Fig. 9.** Response of the system considering steady viscous effects at $U=0.75$ for $L/R=3$, $h/R=0.02$, $n=0,4,5,6$ and $m=1,2$: (a) phase portrait for point ($\xi=1,\theta=\pi/3$); (b) shell deformation shape.

thinness parameter $h/R=0.02$, $n=3$ is the critical mode for $L/R=7$ and $n=5$ for $L/R=3$. Therefore, the solution expansion for $L/R=7$ contains $n=0,2,3$ and 4; and for $L/R=3$ it contains $n=0,4,5$ and 6. Regarding the axial mode content, $m=1$ and 2 is sufficient.

#### 4.2.1. *Case 1: shell with $L/R=3$ and $h/R=0.02$*

For this shell the solution expansion is composed of $n=0,4,5$ and 6 and $m=1$ and 2.

Fig. 4(b) shows the bifurcation diagram of the system with steady viscous forces, and Fig. 4(a) shows the bifurcation diagram for inviscid flow. It is observed that, when fluid viscosity is taken into account, the general character of system behaviour is not much different from the case of inviscid flow, and the system loses stability at $U_{cr}$ still via a supercritical Hopf bifurcation, giving rise to a family of stable periodic orbits (branch B1). At higher flow velocities, the stationary solution undergoes additional supercritical Hopf bifurcations from which other branches of periodic solution emerge. These limit cycles grow in amplitude with increasing flow velocity until they lose stability via a Neimark–Sacker bifurcation (hereafter referred to as a torus bifurcation).

Bifurcation diagrams of the circumferential and the axial deformation of the shell for the case of viscous flow are shown in Figs. 5 and 6, respectively. It is seen that the maximum circumferential displacement is approximately one-fifth of the radial displacement and the maximum axial displacement is about one-eighth of the circumferential one, signifying the importance of in-plane displacements.

Steady viscous forces, however, do modify the dynamics of the system quantitatively. Table 1 presents the values of dimensionless flow velocity at which Hopf bifurcations occur for both viscous and inviscid fluid flow cases. It is seen that steady viscous effects destabilize the stationary solution: that is, the system is subject to flutter at a lower flow velocity. Quantitatively, the critical dimensionless flow velocity of flutter $U_{cr}$ (HB1) becomes approximately 6% smaller when steady viscous forces are taken into account. Also, other Hopf bifurcations occur at a lower flow velocity compared to inviscid flow.

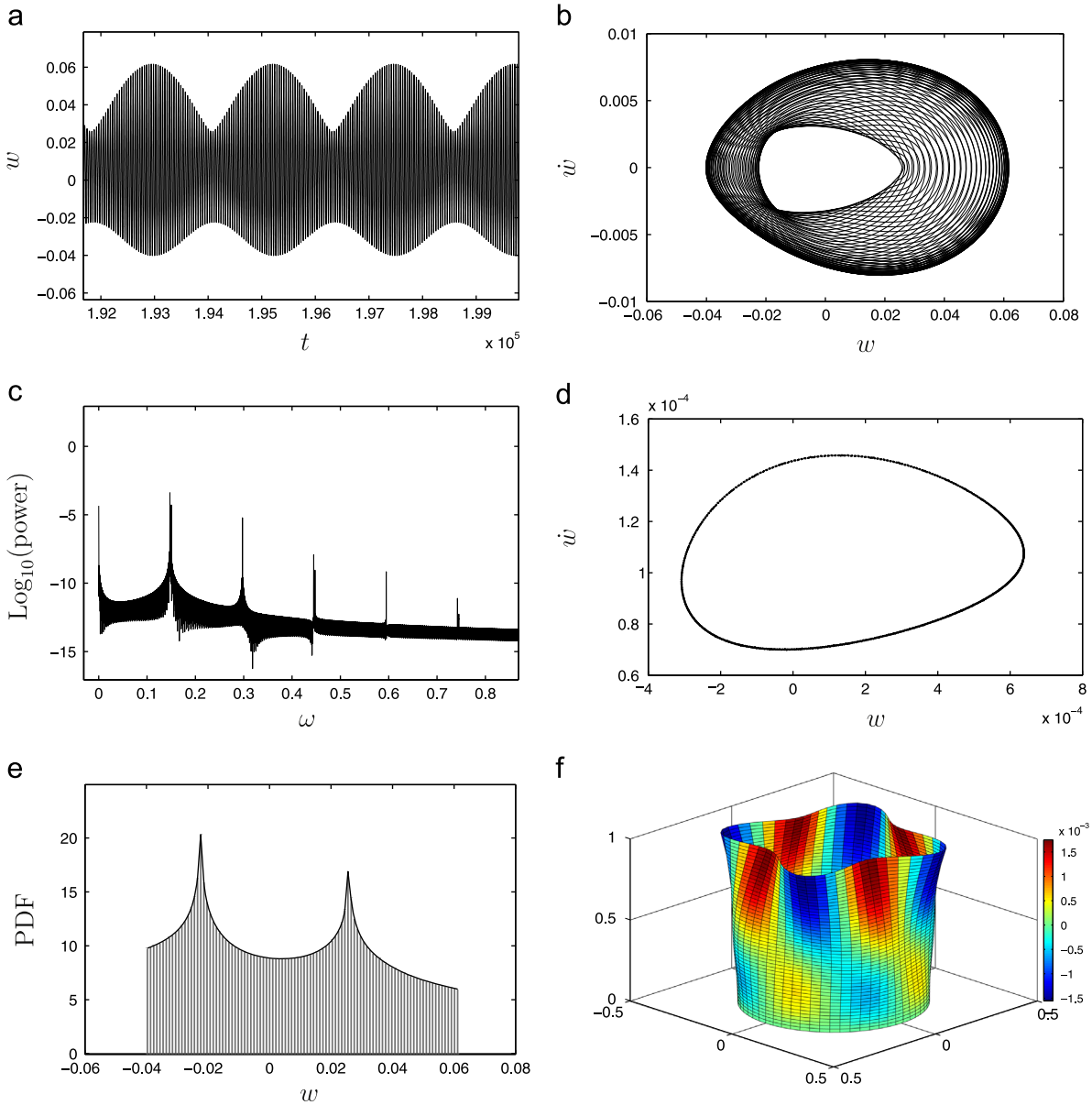



**Fig. 10.** Response of the system considering steady viscous effects at $U=0.804$ for $L/R=3$, $h/R=0.02$, $n=0,4,5,6$ and $m=1,2$: (a) time trace of $w$ ($\xi=1,\theta=\pi/3$); (b) phase portrait; (c) power spectrum; (d) one-sided Poincaré map; (e) PDF; (f) shell deformation shape.

The steady viscous effects, on the other hand, make the stable limit cycles more robust; as a result, they remain stable over a wider range of flow velocity than they do when flow is considered to be inviscid. In other words, these effects regularize the motion. The interval of dimensionless fluid velocity corresponding to the stable part of the branches of periodic orbits for inviscid and viscous flows is given in Table 2. In this table, branch B3 for viscous flow is compared to branch B2 of inviscid flow since they are equivalent according to Fig. 4. It is seen that steady viscous forces broaden the range of stable periodic behaviour of the system. Branch B3 in the bifurcation diagram of inviscid flow is unstable all over; whereas, its equivalent branch for viscous theory (*i.e.*, B2) is stable over some region.

Fig. 7 shows the variation of dimensionless frequency with $U$ of the limit cycles on each branch. It is observed that, in general, the frequency of oscillations, whether stable or unstable, increases (*i.e.*, the period of orbits decreases); however, the family of periodic orbits on B3 always possesses a smaller frequency than those of B1 and B2. The frequency varies almost linearly with $U$.

To better grasp the nature of post-instability behaviour of the system, the time response of the system for $U=0.54$, 0.75, 0.804, 0.84 and 1.0 is computed and analysed. The first two of these velocities each corresponds to a stable periodic response on branch B1 and B3 of Fig. 4(b); $U=0.804$ and 0.84 are in the region where no stable periodic attractor exists, and the last flow velocity, $U=1.0$, falls on the stable region of branch B2.

The response of the system for $U=0.54$ is shown in Fig. 8. The time trace and the phase portrait indicate a periodic motion with a maximum radial amplitude of 0.0058 (in dimensional form $0.29h$), which is located on the stable part of branch B1. The FFT shows a single peak at $\omega=0.16$ and the Poincaré map is a single point, which further testifies that the response is periodic. In addition, the PDF shows the expected two-masted distribution for periodic time signals. A snapshot of the oscillating shell at $t=0.95\times10^5$ is shown in Fig. 8 (f); it is seen that $n=5$ is the predominant circumferential mode in the shell deformation, which due to symmetry travels around the circumference (travelling wave). The information obtained from the time analysis is in accord with the bifurcation diagrams.

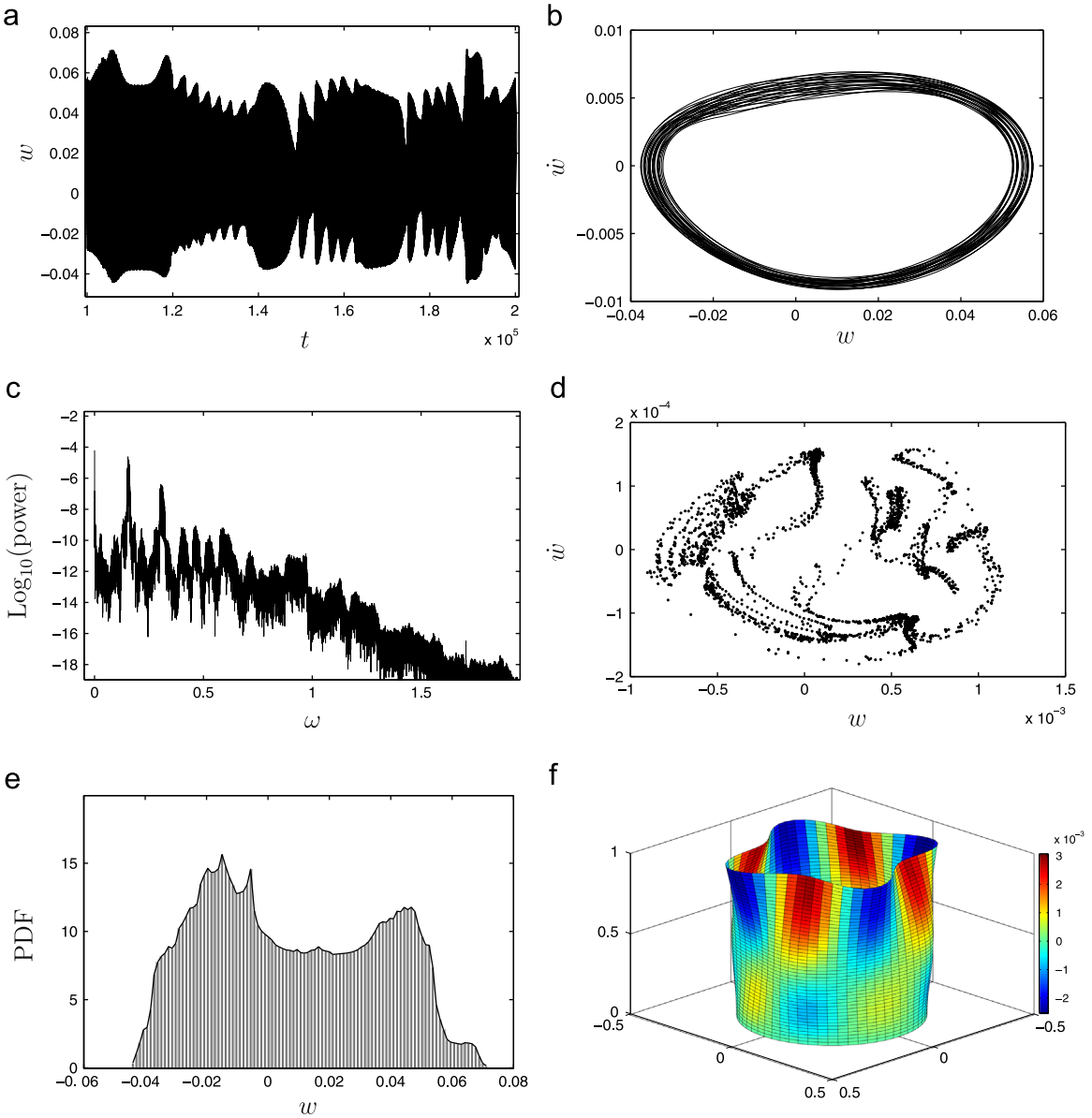



**Fig. 11.** Response of the system considering steady viscous effects at $U=0.84$ for $L/R=3$, $h/R=0.02$, $n=0,4,5,6$ and $m=1,2$: (a) time trace of $w$ ($\xi=1,\theta=\pi/3$); (b) phase portrait; (c) power spectrum; (d) one-sided Poincaré map; (e) PDF; (f) shell deformation shape.

The system behaviour for $U=0.75$ is shown in Fig. 9. The phase portrait shows a periodic motion with a maximum radial amplitude of 0.0352 (in dimensional form $1.76h$); this stable limit cycle, having a main frequency of $\omega=0.14$, is located on branch B3 with predominant circumferential mode of $n=4$ [see Fig. 9(b)]. The solution at this velocity is still of travelling wave type. The Poincaré map and PDF of the response have the same properties as those for $U=0.54$. It is noticed that for $U=0.75$ the oscillation has a larger amplitude and a different shape (*i.e.*, a mode switch has occurred.).

The family of stable periodic orbits on branch B3 loses stability via a torus bifurcation at $U=0.8$ and regains stability at $U=0.849$; in the interval $(0.8, 0.849)$, no stable periodic attractor exists on other branches, either. Hence, it is of interest to know how the system behaves immediately after stable periodic attractors cease to exist. To this end, the system response at $U=0.804$ is obtained and analysed (Fig. 10). The time trace is amplitude-modulated with $\max(w)=3.08h$ and the phase portrait fills a certain region of phase space, both of which imply a quasiperiodic motion. In addition, we may ascertain that the response is quasiperiodic through the FFT which shows two adjacent incommensurate peaks ($\omega_1=0.1473$ and $\omega_2=0.1501$) and their superharmonics, and also through the Poincaré map which is a closed curve (*i.e.*, the motion trajectory is on a 2-torus, topologically). The predominant circumferential mode is still $n=4$ and the motion is non-stationary along the circumference [see Fig. 10(f)]. The PDF, shown in Fig. 10(e), suggests that the motion trajectory now visits a wider region of the phase space.

Farther in the unstable region, the response of the system for $U=0.84$ is shown in Fig. 11. The time trace indicates an "almost" quasiperiodic response (on a recently disintegrated torus) with bursts of chaos. In fact, for the set of initial conditions used, the motion trajectory bounces around the ghost of a destroyed quasiperiodic orbit for some period of time (laminar phase) until it is attracted to a chaotic attractor, and after some time (chaotic phase) reinjected again into the neighbourhood of the ghost. This type of behaviour is a kind of transient chaos and, of course, it is due to global bifurcations in the system. The phase portrait, plotted mostly in the laminar phase, shows a not too irregular motion with a maximum amplitude of $3.58h$. The FFT contains a main frequency at $\omega=0.1534$ with some broadband regions, and the Poincaré map is a cloud of points with a fractal structure, indicating a strange attractor. Fig. 11(f) suggests that $n=4$ is still the predominant circumferential mode.

To further investigate the characteristics of system behaviour at $U=0.84$, the four largest Lyapunov exponents have been computed [Fig. 12(a)]; it is seen that, after convergence, two out of four exponents are positive with small amplitudes (responsible for a weak chaotic behaviour) and two are almost zero (indication of a quasiperiodic motion). The autocorrelation of the response, shown in Fig. 12(b), indicates that the states of the system become less correlated as time increases but with modulation due to quasiperiodic components.

Finally, the system response for $U=1.0$ is investigated. From the bifurcation diagram in Fig. 4(b), the response should be a stable limit cycle on branch B2. Fig. 13 shows the response of the system at $U=1$. The phase portrait indicates a periodic motion with a maximum amplitude of $1.28h$. At this velocity, the shell oscillates predominantly at $n=6$ [Fig. 13(b)], and the motion is a travelling wave.

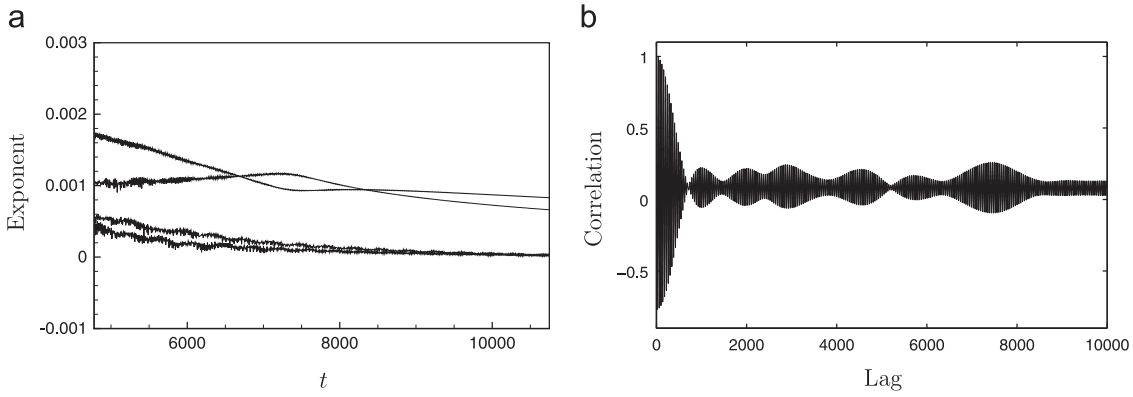



**Fig. 12.** Response of the system considering steady viscous effects at $U=0.84$ for $L/R=3$, $h/R=0.02$, $n=0,4,5,6$ and $m=1,2$: (a) Lyapunov exponents; (b) autocorrelation of the time response.

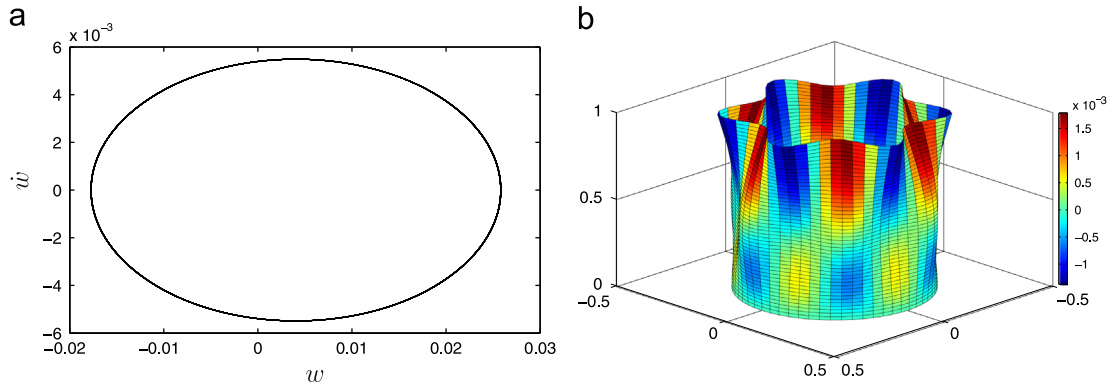



**Fig. 13.** Response of the system considering steady viscous effects at $U=1.0$ for $L/R=3$, $h/R=0.02$, $n=0,4,5,6$ and $m=1,2$: (a) phase portrait for point $(\xi=1,\theta=\pi/3)$; (b) shell deformation shape.

*4.2.2. Case 2: shell with $L/R=7$ and $h/R=0.02$*

For this geometry, the bifurcation diagrams of the system with and without steady viscous forces are shown in Fig. 14(b) and (a), respectively. In these diagrams the maximum radial deformation of the shell is used as a measure and the dimensionless flow velocity as the bifurcation parameter. It is observed that in both cases the system loses stability through a supercritical Hopf bifurcation.

In the post-instability region of the system with viscous flow [Fig. 14(b)], the first two Hopf bifurcations give rise to periodic solutions which are stable over a range of flow velocity and sometimes even overlapping ranges (meaning that two stable periodic orbits coexist). The stable periodic solution emerging from the first Hopf bifurcation (on branch B1) grows in amplitude and frequency [see Fig. 17] as the flow velocity is increased, until it becomes unstable via a torus bifurcation at $U=0.723$ and continues to exist as an unstable branch. The second branch, denoted by B2, starts as an unstable solution, then gains stability at $U=0.401$ and finally becomes unstable through a torus bifurcation at $U=0.703$. It is noted that the transition to stability or instability should arise through a bifurcation, which is sometimes difficult to catch numerically (requiring very small steps in the computation). No stable periodic orbits have been found for the third Hopf bifurcation.

Bifurcation diagrams of the circumferential component ($v$) and the axial component ($u$) of the shell deformation are shown in Figs. 15 and 16, respectively. It is observed that for all periodic orbits, over the range of flow velocity considered, the shell deformation is the largest in the radial direction and the smallest

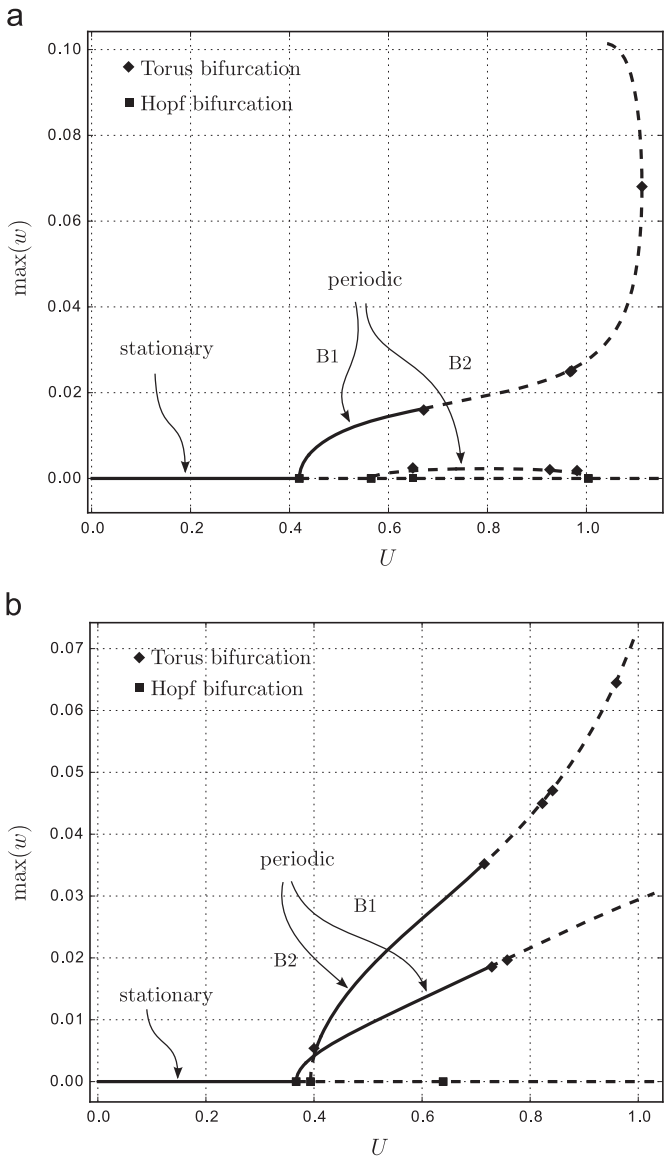



**Fig. 14.** Maximum value of the dimensionless radial deformation $w$ at $(\xi=1,\theta=\pi/3)$ versus dimensionless flow velocity $U$; ——, stable solution; – – – – –, unstable solution; B1, branch 1, and so on; $L/R=7$, $h/R=0.02$; $n=0,2,3,4$ and $m=1,2$; (a) inviscid flow; (b) with steady viscous effect.

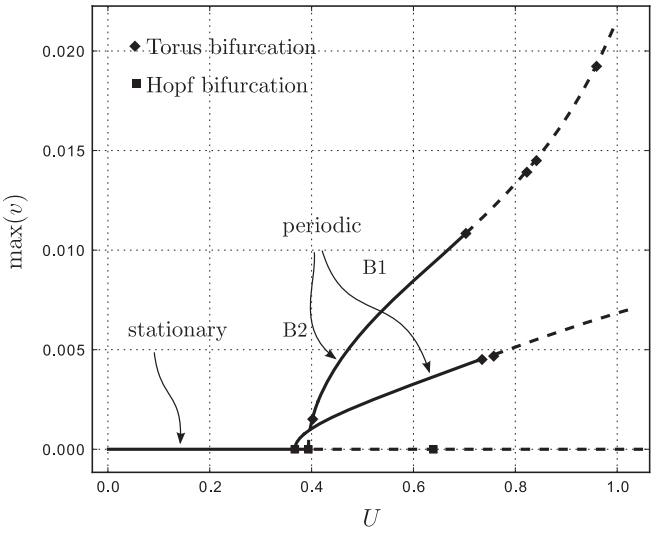



**Fig. 15.** Maximum value of the dimensionless circumferential deformation $v$ at $(\xi=1,\theta=\pi/3)$ versus dimensionless flow velocity $U$ for the system considering steady viscous effects; ——, stable solution; – – – – –, unstable solution; B1, branch 1, and so on; $L/R=7$, $h/R=0.02$; $n=0,2,3,4$ and $m=1,2$.

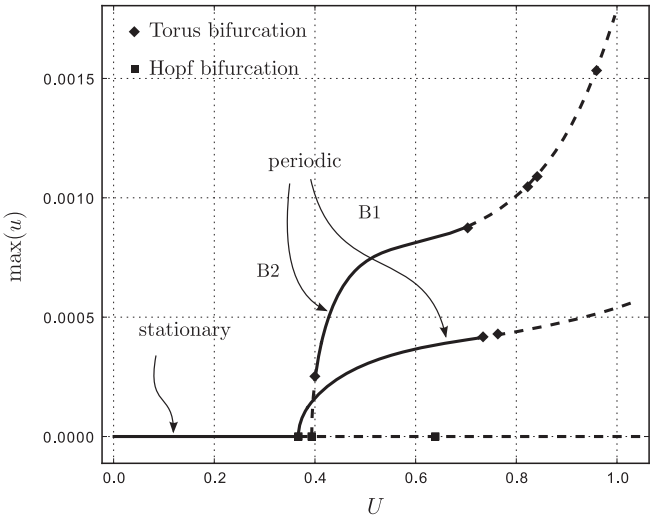



**Fig. 16.** Maximum value of the dimensionless axial deformation $u$ at $(\xi=1,\theta=\pi/3)$ versus dimensionless flow velocity $U$ for the system considering steady viscous effects; ——, stable solution; – – – – –, unstable solution; B1, branch 1, and so on; $L/R=7$, $h/R=0.02$; $n=0,2,3,4$ and $m=1,2$.

**Table 3**
Dimensionless flow velocities associated with Hopf bifurcations of inviscid and viscous theory, $L/R=7$.

| Flow | $HB_1$ ($U_{cr}$) | $HB_2$ | $HB_3$ |
|---|---|---|---|
| Inviscid | $U_{inv}=0.419$ | 0.565 | 0.648 |
| Viscous | $U_{vis}=0.367$ | 0.393 | 0.639 |
| $(U_{vis}-U_{inv})/U_{inv}$ | $\sim -12\%$ | $\sim -30\%$ | $\sim -1\%$ |

**Table 4**
The interval of $U$ for stable periodic orbits on branches B1, B2 and B3, predicted by inviscid and viscous theories (cf. Fig. 14), $L/R=7$.

| Flow | Branch B1 | Branch B2 |
|---|---|---|
| Inviscid | $I_{inv}=(0.419, 0.671)$ | Unstable |
| Viscous | $I_{vis}=(0.367, 0.723)$ | (0.401, 0.703) |
| $(\lvert I_{vis}\rvert-\lvert I_{inv}\rvert)/\lvert I_{inv}\rvert$ | $\sim 41\%$ | – |

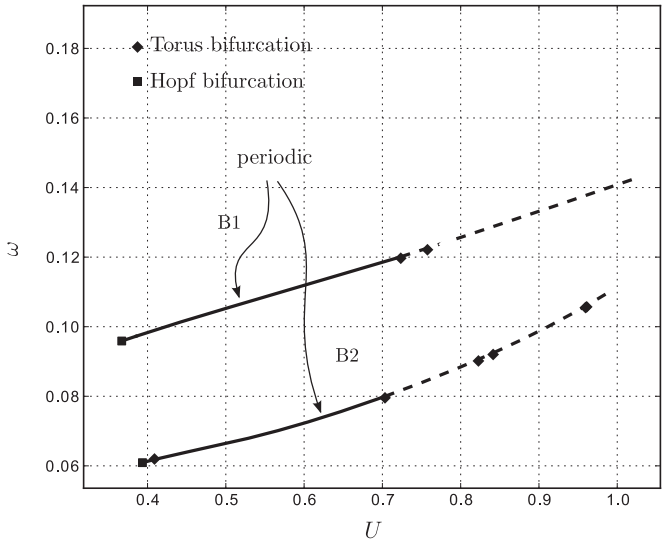



**Fig. 17.** Dimensionless frequency of limit cycles versus dimensionless flow velocity $U$ for the system considering steady viscous effects; ——, stable solution; – – – – –, unstable solution; B1, branch 1, and so on; $L/R=7$, $h/R=0.02$; $n=0,2,3,4$ and $m=1,2$.

in the axial direction. The shell deformation in the circumferential direction is approximately one-third of that in the radial direction.

Although taking fluid viscosity into account does not make a qualitative change in the general behaviour of the system, it does alter the system quantitatively. The values of dimensionless flow velocity at which Hopf bifurcations occur on the stationary solution are listed in Table 3 for both viscous and inviscid flows. It is seen that when the flow is viscous the Hopf bifurcations occur at lower flow velocities than their counterparts in inviscid flow; in other words, the steady viscous forces destabilize the system (*i.e.*, the system flutters at a lower flow velocity). In particular, the critical flow velocity of flutter ($U_{cr}$) has decreased by $\sim 12\%$.

Steady viscous forces, however, extend the interval of flow velocity over which the periodic behaviour of the system is stable. That is, the stable limit cycles last longer in the presence of viscosity. Table 4 shows the interval of dimensionless flow velocities over which branches B1, B2 and B3 are stable for both inviscid and viscous flows. It is seen that, except for B1, the stable limit cycles on the other two branches have been stabilized. Particularly, branch B2, unstable all over its range in the case of inviscid flow, gains stability over a considerable range when viscosity is taken into account. Fig. 17 shows how the dimensionless frequency of the limit cycles on each branch changes with $U$. It is observed that the frequency of oscillation increases; however, the family of periodic orbits on B2 has smaller frequency than those of B1. The frequency varies almost linearly with $U$.

To better understand the nature of the post-instability behaviour of the system with steady viscous forces taken into account, time responses of the system for velocities $U=0.6$, 0.73, and 0.75 have been obtained and analysed. For dimensionless flow velocity $U=0.6$ there are stable periodic orbits on both B1 and B2; given appropriate initial conditions, the solution may land on either of them. Based on the bifurcation diagrams, there are no stable periodic attractors after $U=0.723$; thus, solutions at $U=0.73$ and 0.75 will be aperiodic.

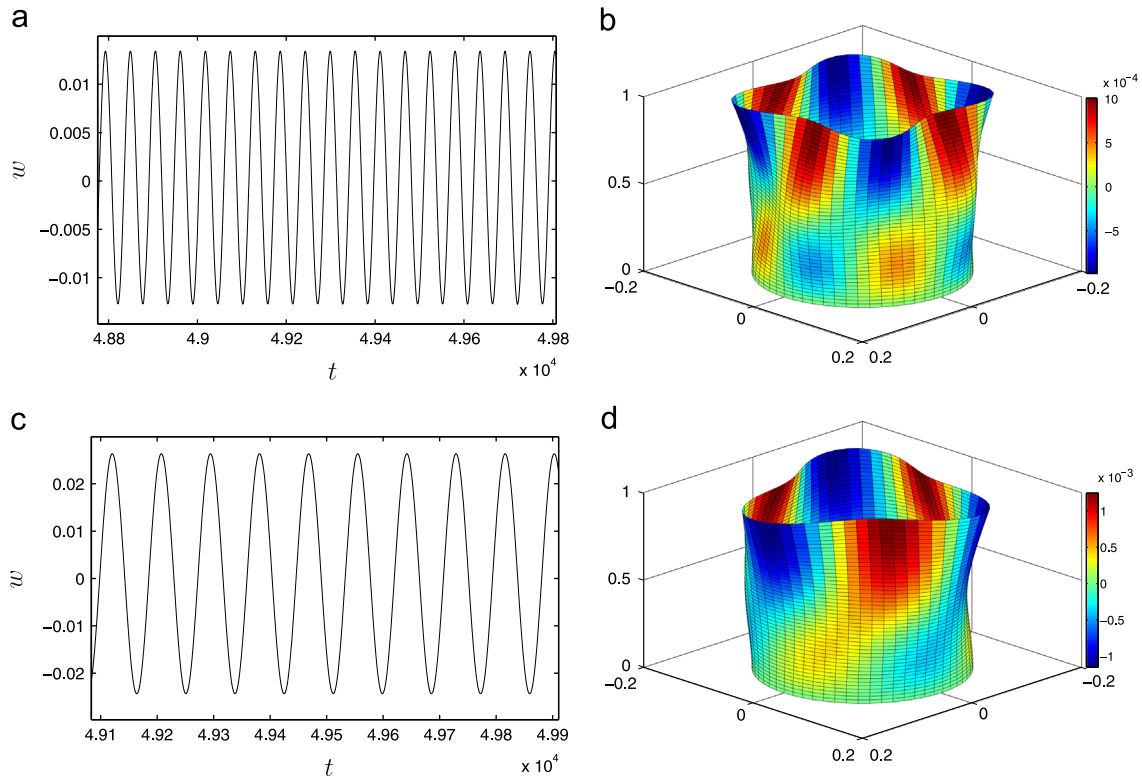



**Fig. 18.** Response of the system considering steady viscous effects at $U=0.6$ for $L/R=7$, $h/R=0.02$, $n=0,2,3,4$ and $m=1,2$: (a) time trace of $w$ ($\xi=1, \theta=\pi/3$) on branch B1; (b) shell deformation shape on branch B1; (c) time trace of $w$ ($\xi=1, \theta=\pi/3$) on branch B2; (d) shell deformation shape on branch B2.

The response of the system at $U=0.60$ is shown in Fig. 18. Two sets of initial conditions have been utilized; one leading to a periodic orbit on branch B1, shown in Fig. 18(a) and (b), and the other to a periodic orbit on branch B2, shown in Fig. 18(c) and (d). The solution on B1 has a maximum amplitude of $0.67h$ in the radial direction with $n=4$ being the predominant circumferential mode, while the solution on B2 has a larger amplitude ($1.31h$) in which $n=3$ is the predominant circumferential mode. In both cases, the solution is a travelling wave around the shell circumference.

Next, $U=0.73$ is considered in order to investigate what occurs right after the stable periodic orbits cease to exist. Fig. 19 shows the response of the system. The time trace resembles an amplitude-modulated period-3 orbit and the phase portrait, shown in a 3-D subspace, fills up a certain region of the phase space; it is deduced that the motion is quasiperiodic, with $\max(w)=0.1207$ (in dimensional form $6.03h$). In the frequency spectrum, shown in Fig. 19(c), three distinct peaks may be recognized while there are many superharmonics. The three closed curves in the Poincaré map attest to the response being quasiperiodic. This type of motion is, in fact, very interesting because it is a quasiperiodic orbit around an unstable period-3 limit cycle. The period-3 orbit itself may have been due a phase-locking of its predecessor quasiperiodic orbit. The PDF possesses three relatively sharp peaks (a legacy of the period-3 orbit), and the shell deformation shape shows that $n=2$ is now the predominant circumferential mode.

Fig. 20 shows the response of the system at $U=0.75$; from the time trace and the phase portrait, it is deduced that the motion is aperiodic and has a maximum amplitude of $8.03h$ in the radial direction. The broadband spectrum, shown in Fig. 20(c), and the cloud of points in the Poincaré map attest to the response being chaotic. The PDF looks like a bimodal distribution, and the shell deformation shape shows that $n=2$ is still the predominant circumferential mode. To analyse the chaoticity of the behaviour further, the four largest Lyapunov exponents are computed and shown in Fig. 21(a); it is seen that, after convergence, three out of four exponents are clearly positive, albeit not very large. The autocorrelation of the response, shown in Fig. 21(b), indicates that the system loses its memory, gradually; however, the presence of modulated portions imply the existence of some quasi-periodic component in the response.

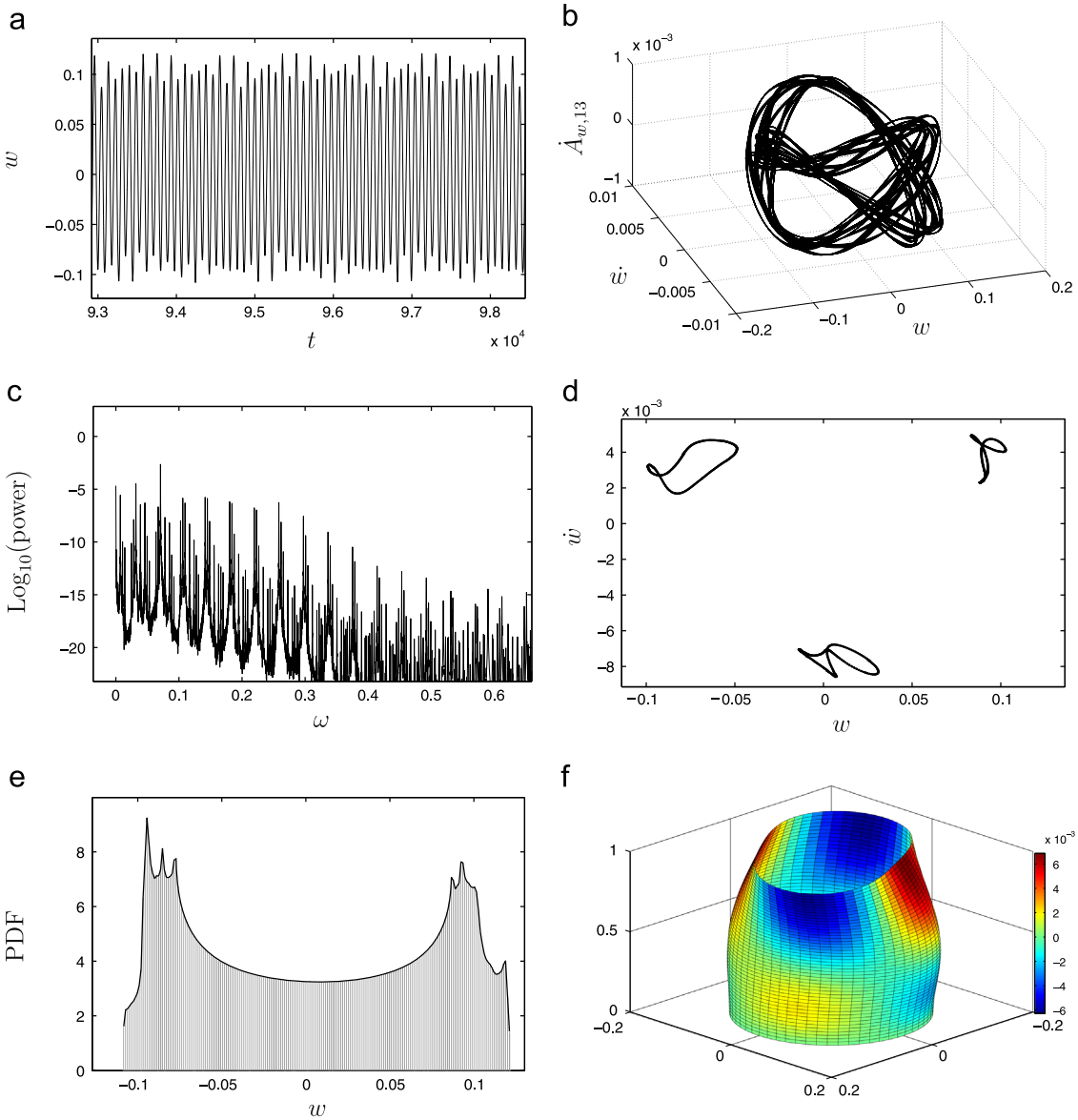



**Fig. 19.** Response of the system considering steady viscous effects at $U=0.73$ for $L/R=7$, $h/R=0.02$, $n=0,2,3,4$ and $m=1,2$: (a) time trace of $w$ ($\xi=1,\theta=\pi/3$); (b) phase portrait; (c) power spectrum; (d) one-sided Poincaré map; (e) PDF; (f) shell deformation shape.

## 5. Concluding remarks

The non-linear dynamics and stability of cantilevered circular cylindrical shells conveying fluid are examined by taking into account the steady viscous forces. The effect of such forces on the non-linear behaviour of this system has not been studied before.

Another contribution of this study is the inclusion of axisymmetric modes ($n=0$) in the solution expansion. If these modes are

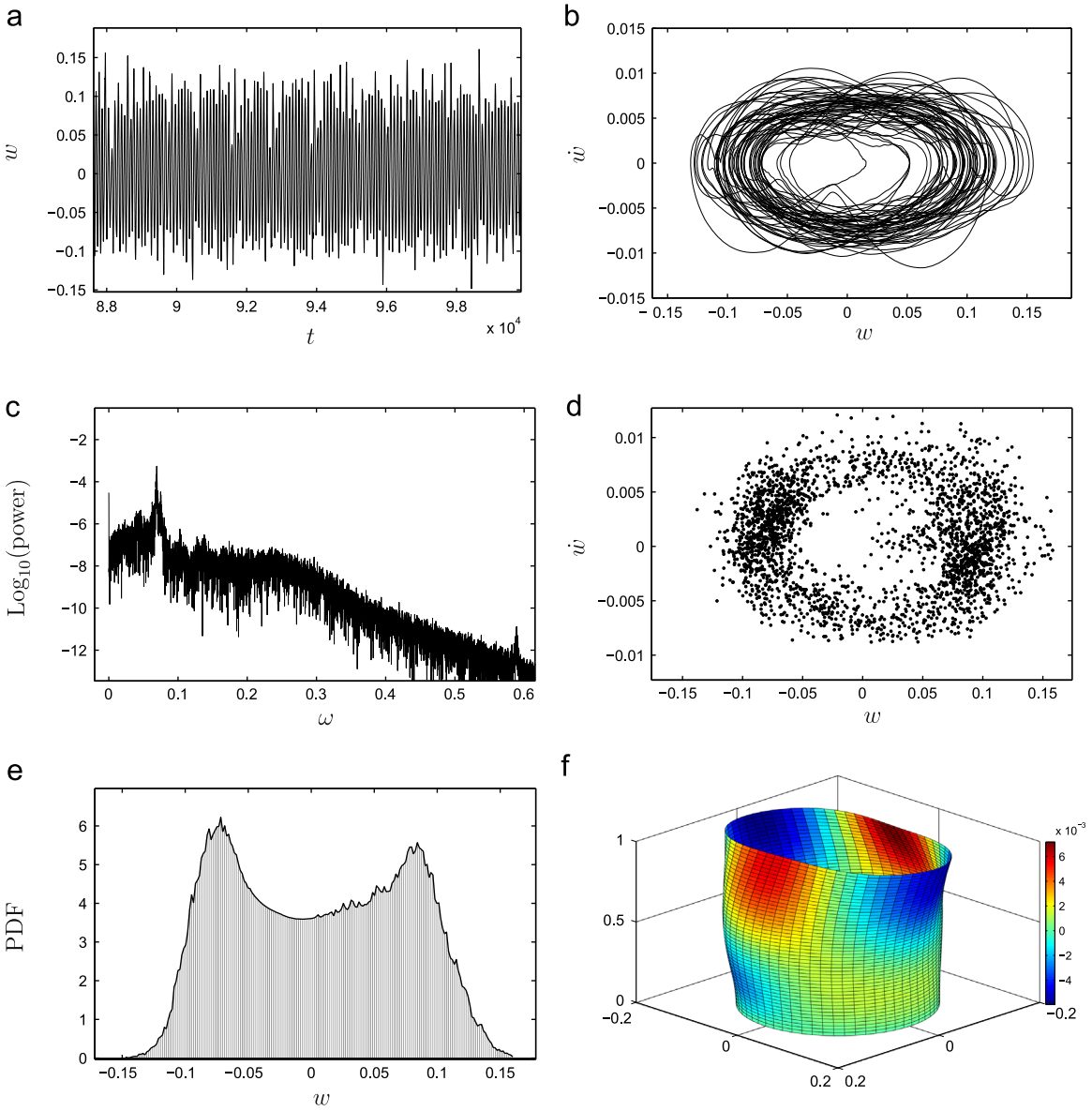



**Fig. 20.** Response of the system considering steady viscous effects at $U=0.75$ for $L/R=7$, $h/R=0.02$, $n=0,2,3,4$ and $m=1,2$: (a) time trace of $w$ ($\xi=1,\theta=\pi/3$); (b) phase portrait; (c) power spectrum; (d) one-sided Poincaré map; (e) PDF; (f) shell deformation shape.

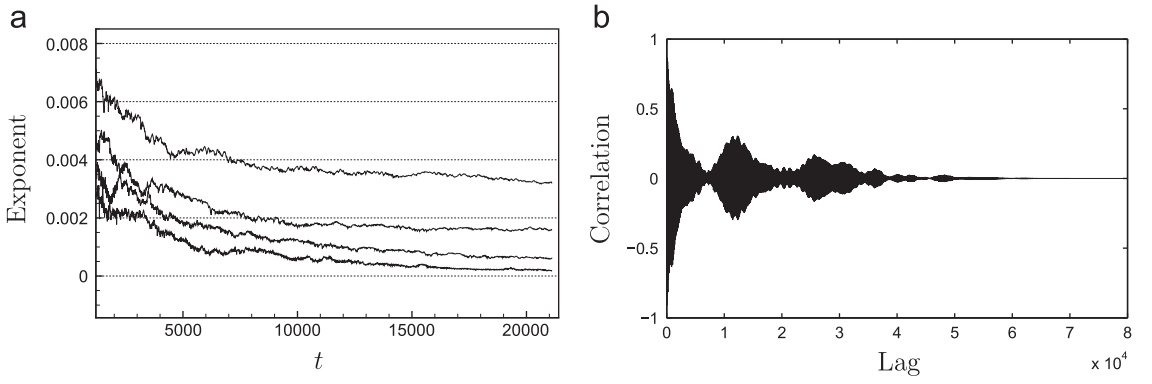



**Fig. 21.** Response of the system considering steady viscous effects at $U=0.75$ for $L/R=3$, $h/R=0.02$, $n=0,2,3,4$ and $m=1,2$: (a) Lyapunov exponents; (b) autocorrelation of the time response.

omitted, the system becomes harder at large amplitudes due to excessive mid-surface stretching, although the qualitative response of the system does not alter much (cf. [15]). The added mass associated with $n=0$ modes is obtained by assuming that the fluid is still and contained by the shell (the added mass does not depend on the fluid velocity), while the Coriolis and centrifugal forces are found by assuming a zero outflow length.

It is found that the steady viscous forces do not qualitatively change the general dynamics of the system; the system still loses stability by flutter. The stable static configuration becomes unstable through a supercritical Hopf bifurcation giving rise to stable periodic oscillations (limit cycles) the amplitude of which grows with flow velocity until they become unstable via a torus bifurcation and the oscillations become aperiodic (*i.e.*, quasiperiodic or chaotic).

The steady viscous effects, however, change the system quantitatively. Counter-intuitively, the stationary solution becomes unstable at a lower flow velocity than when such effects are neglected; *i.e.*, viscosity in the flow destabilizes the system. Also, the additional Hopf bifurcations on the unstable stationary solution occur at a lower flow velocity.

When the frictional forces due to the fluid viscosity are taken into account, the system is not merely circulatory [27] but rather non-conservative in a more general sense; it is known that, for non-conservative systems, such as this one, distributed axial forces may have a destabilizing effect.

The steady viscous forces, on the other hand, make the stable limit cycles more robust; in other words, these forces expand the range of flow velocity over which the periodic solutions are stable. Furthermore, the amplitude of the limit cycles is now slightly larger most probably because of the fluid pressurization within the shell.

The non-linear terms due to curvature changes and twist proved to have very little effect on the non-linear response of the system within the range of deformation of the order of the shell thickness. These terms, perhaps, become more important for large rotations.

The stable periodic responses of the system are of the travelling wave type, due to axial symmetry, and their frequency of oscillation increases with the flow velocity. In the post-instability region, the system exhibits rich dynamics and follows a quasiperiodic route to chaos. It is seen from the Lyapunov exponents that the chaotic responses become more erratic as the flow velocity is increased.

## Acknowledgements

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## Appendix A. Stresses and strains

The strain displacement relationships, based on Flügge's shell theory, are

$$\varepsilon_x = u_{,x} - z w_{,xx} + \varepsilon_x^{(2)},$$
$$\varepsilon_y = v_{,y} - \frac{R}{R-z} z w_{,yy} + \varepsilon_y^{(2)},$$
$$\gamma_{xy} = \frac{R-z}{R} v_{,x} + \frac{R}{R-z} u_{,y} - \left(1+\frac{R}{R-z}\right) z w_{,xy} + \gamma_{xy}^{(2)}, \tag{A.1}$$

where the non-linear terms of the strain–displacement relationships, designated by the superscript (2), may be expressed as

$$\varepsilon_x^{(2)} = \varepsilon_{x0}^{(2)} - z\kappa_{x0}^{(2)}, \quad \varepsilon_y^{(2)} = \varepsilon_{y0}^{(2)} - \frac{z}{R-z}\kappa_{y0}^{(2)},$$
$$\gamma_{xy}^{(2)} = \gamma_{xy0}^{(2)} - z\left(\kappa_{1xy0}^{(2)} + \frac{1}{R-z}\kappa_{2xy0}^{(2)}\right),$$

in which the non-linear terms of mid-surface stretching are given by

$$\varepsilon_{x0}^{(2)} = \tfrac{1}{2}(u_{,x}^2 + v_{,x}^2 + w_{,x}^2),$$
$$\varepsilon_{y0}^{(2)} = \frac{1}{2}\left[u_{,y}^2 + \left(v_{,y} - \frac{w}{R}\right)^2 + \left(w_{,y} + \frac{v}{R}\right)^2\right],$$
$$\gamma_{xy0}^{(2)} = \frac{1}{2}\left[u_{,x}u_{,y} + v_{,x}\left(v_{,y} - \frac{w}{R}\right) + w_{,x}\left(w_{,y} - \frac{v}{R}\right)\right]; \tag{A.2}$$

and the non-linear terms of curvature changes and twist are written as

$$\kappa_{x0}^{(2)} = \frac{1}{R}v_{,x}^2 + v_{,x}w_{,xy} + u_{,x}w_{,xx},$$
$$\kappa_{y0}^{(2)} = \frac{1}{R}v w_{,y} + w_{,y}^2 - w w_{,yy} + R v_{,y} w_{,yy} + R u_{,y} w_{,xy},$$
$$\kappa_{1xy0}^{(2)} = w_{,y}v_{,x} + v_{,y}w_{,xy},$$
$$\kappa_{2xy0}^{(2)} = w_{,y}w_{,x} - w w_{,xy} + R u_{,x} w_{,xy} + R u_{,y} w_{,xx}. \tag{A.3}$$

From Hooke's law, it follows that

$$\sigma_x = \frac{E}{1-\nu^2}(\varepsilon_x + \nu\varepsilon_y), \quad \sigma_y = \frac{E}{1-\nu^2}(\varepsilon_y + \nu\varepsilon_x),$$
$$\tau_{xy} = \frac{E}{2(1+\nu)}\gamma_{xy}. \tag{A.4}$$

## Appendix B. Contribution of non-linear terms of curvature changes and twist

Modifying expressions to $L_i$, $i=1,2,3$, denoted by $L_i^C$, are defined by

$$L_1^C = -[M_x w_{,xx} + M_{yx} w_{,xy}]_{,x} - [M_{yx} w_{,xx} + M_y w_{,xy}]_{,y},$$
$$L_2^C = \frac{1}{R^2}M_y w_{,y} - \left[\frac{2}{R}M_x v_{,x} + M_x w_{,xy} + M_{xy} w_{,yy}\right]_{,x} - [M_{xy} w_{,xy} + M_y w_{,yy}]_{,y},$$
$$\begin{aligned} L_3^C = {} & -\frac{1}{R}(M_{yx} w_{,xy} + M_y w_{,yy}) - \left[\frac{1}{R}M_{yx} w_{,y}\right]_{,x} \\ & - \left[\frac{1}{R^2}(v + 2R w_{,y}) + \frac{1}{R}M_{yx} w_{,x}\right]_{,y} + [M_x u_{,x} + M_{yx} u_{,y}]_{,xx} \\ & + \left[-\frac{1}{R}M_{yx} w + M_{yx} u_{,x} + M_y u_{,y} + M_x v_{,x} + M_{xy} v_{,y}\right]_{,xy} \\ & + \left[-\frac{1}{R}M_y w + M_{xy} v_{,x} + M_y v_{,y}\right]_{,yy}, \end{aligned} \tag{B.1}$$

and modifying expressions to $R_i$, $i=1,2,3,4$, denoted by $R_i^C$, are given by

$$R_1^C = -(M_x w_{,xx} + M_{yx} w_{,xy}),$$
$$R_2^C = -\left(\frac{1}{R}M_x v_{,x} + M_x w_{,xy} + M_{xy} w_{,yy}\right),$$
$$\begin{aligned} R_3^C = {} & -\frac{1}{R}M_{yx} w_{,y} + [M_x u_{,x} + M_{yx} u_{,y}]_{,x} \\ & + \left[-\frac{1}{R}M_{yx} w + M_{yx} u_{,x} + M_y u_{,y} + M_x v_{,x} + M_{xy} v_{,y}\right]_{,y}, \end{aligned}$$
$$R_4^C = -(M_x u_{,x} + M_{yx} u_{,y}). \tag{B.2}$$

The contribution of non-linear terms of curvature changes and twist to the stress resultants and stress couples, denoted by superscript $C$, are

$$N_x^C = \frac{\mathcal{D}}{R}\kappa_{x0}^{(2)}, \quad N_y^C = -\frac{\mathcal{D}}{R^2}\kappa_{y0}^{(2)},$$
$$N_{xy}^C = \frac{(1-\nu)\mathcal{D}}{2R}\kappa_{1xy0}^{(2)}, \quad N_{yx}^C = -\frac{(1-\nu)\mathcal{D}}{2R^2}\kappa_{2xy0}^{(2)},$$

$$M_x^C = -\mathcal{D}\left(\kappa_{x0}^{(2)} + \frac{\nu}{R}\kappa_{y0}^{(2)}\right), \quad M_y^C = -\mathcal{D}\left(\nu\kappa_{x0}^{(2)} + \frac{1}{R}\kappa_{y0}^{(2)}\right),$$
$$M_{xy}^C = -\frac{(1-\nu)\mathcal{D}}{2}\left(\kappa_{1xy0}^{(2)} + \frac{1}{R}\kappa_{2xy0}^{(2)}\right), \quad M_{yx}^C = M_{xy}^C, \tag{B.3}$$

where $\mathcal{D} = Eh^3/12(1-\nu_s)$ (flexural rigidity).

## Appendix C. Virtual work due to unsteady fluid-dynamic forces

The dimensionless form the virtual work done by the unsteady fluid-dynamic forces is given by

$$\delta\mathcal{W}_f = \sum_{j=1}^{\infty}\sum_{m=1}^{\infty}\left(-\frac{\gamma}{\varepsilon\beta}I_{jm0}^{(1)}\frac{\partial^2}{\partial t^2}A_{w,m0} + \frac{\gamma}{\varepsilon}I_{jm0}^{(2)}U\frac{\partial}{\partial t}A_{w,m0} + \frac{\gamma\beta}{\varepsilon}I_{jm0}^{(3)}U^2A_{w,m0}\right)$$
$$\times\delta A_{w,j0} + \frac{1}{2}\sum_{n=1}^{\infty}\sum_{j=1}^{\infty}\sum_{m=1}^{\infty}\left[\left(-\frac{\gamma}{\varepsilon\beta}I_{jmn}^{(1)}\frac{\partial^2}{\partial t^2}A_{w,mn}\right.\right.$$
$$\left. + \frac{\gamma}{\varepsilon}I_{jmn}^{(2)}U\frac{\partial}{\partial t}A_{w,mn} + \frac{\gamma\beta}{\varepsilon}I_{jmn}^{(3)}U^2A_{w,mn}\right)\delta A_{w,jn}.$$
$$\left. + \left(-\frac{\gamma}{\varepsilon\beta}I_{jmn}^{(1)}\frac{\partial^2}{\partial t^2}B_{w,mn} + \frac{\gamma}{\varepsilon}I_{jmn}^{(2)}U\frac{\partial}{\partial t}B_{w,mn} + \frac{\gamma\beta}{\varepsilon}I_{jmn}^{(3)}U^2B_{w,mn}\right)\delta B_{w,jn}\right]. \tag{C.1}$$

In the above equation, the first term in the coefficient of either $\delta A_{w,jn}$ or $\delta B_{w,jn}$ represents the added mass, the second term Coriolis forces and the last one centrifugal forces. $I_{jmn}^{(\cdot)}$ are the dimensionless integrals defined by[1]

$$I_{jm0}^{(1)} = \sum_{i=1}^{\infty} F_{mi}F_{ji}\frac{4\mathrm{I}_0(i\pi\beta)}{i\mathrm{I}_0'(i\pi\beta)},$$
$$I_{jmn}^{(1)} = \int_{-\infty}^{+\infty}\frac{\mathrm{I}_n(\alpha\beta)}{\alpha\mathrm{I}_n'(\alpha\beta)}[H_{jm}(\alpha)+N_{jm}(\alpha)]\,\mathrm{d}\alpha, \quad n=1,2,\ldots$$
$$I_{jmn}^{(2)} = \int_{-\infty}^{+\infty}\frac{2i\mathrm{I}_n(\alpha\beta)}{\mathrm{I}_n'(\alpha\beta)}[H_{jm}(\alpha)+N_{jm}(\alpha)]\,\mathrm{d}\alpha, \quad n=0^1,1,2,\ldots$$
$$I_{jmn}^{(3)} = \int_{-\infty}^{+\infty}\frac{\alpha\mathrm{I}_n(\alpha\beta)}{\mathrm{I}_n'(\alpha\beta)}[H_{jm}(\alpha)+N_{jm}(\alpha)]\,\mathrm{d}\alpha, \quad n=0^1,1,2,\ldots, \tag{C.2}$$

where $F_{mi} = \int_0^1 \phi_m(\xi)\sin(i\pi\xi)\,\mathrm{d}\xi$, $H_{jm}(\alpha) = H_j(-\alpha)H_m(\alpha)$, and $N_{jm}(\alpha) = H_j(-\alpha)N_m(\alpha)$ in which

$$H_j(\alpha) = \int_0^1 \phi_j(\xi)\mathrm{e}^{i\alpha\xi}\,\mathrm{d}\xi, \quad N_j(\alpha) = \int_1^{\infty} R_j(\xi)\mathrm{e}^{i\alpha\xi}\,\mathrm{d}\xi. \tag{C.3}$$

On physical grounds, the flow perturbations die out due to dissipation, and the emanating jet disintegrates some finite distance beyond the shell end. Therefore, in the numerical calculations, the upper bound of the second integral in (C.3), *i.e.*, $\infty$, is replaced by $l=5$ which is physically sound and it guarantees convergence; this means that the outflow length over which the perturbations die out is $(l-1)$ or four times the shell length.

The integrals (C.2) are calculated numerically using the Mathematica software [28] with an adaptive step-size algorithm. The integration range $(-\infty,\infty)$ is approximated by $(-1000, 1000)$, which proved sufficient to yield convergence. Furthermore, in the numerical computation of $I_{jm0}^{(1)}$, in order to reach convergence, 300 is used as the upper bound of summation.

[1] For the case of axisymmetric modes ($n=0$) the outflow length must be set to zero, *i.e.*, $l=1$, which implies that $N_j(\alpha)\equiv 0$.

## References


[1] M.P. Païdoussis, Fluid–Structure Interactions, Slender Structures and Axial Flow, vol. 2, Elsevier Academic Press, London, 2004.
[2] M.P. Païdoussis, J.P. Denise, Flutter of thin cylindrical shells conveying fluid, Journal of Sound and Vibration 20 (1972) 9–26.
[3] D.S. Weaver, T.E. Unny, On the dynamic stability of fluid-conveying pipes, Journal of Applied Mechanics 40 (1973) 48–52.
[4] L.K. Shayo, C.H. Ellen, The stability of finite length circular cross-section pipes conveying inviscid fluid, Journal of Sound and Vibration 37 (1974) 535–545.
[5] L.K. Shayo, C.H. Ellen, Theoretical studies of internal flow-induced instabilities of cantilever pipes, Journal of Sound and Vibration 56 (1978) 463–474.
[6] M.P. Païdoussis, A.K. Misra, S.P. Chan, Dynamics and stability of coaxial cylindrical shells conveying viscous fluid, Journal of Applied Mechanics 52 (1985) 389.
[7] M.P. Païdoussis, V.B. Nguyen, A.K. Misra, A theoretical study of the stability of cantilevered coaxial cylindrical shells conveying fluid, Journal of Fluids and Structures 5 (1991) 127–164.
[8] V.B. Nguyen, M.P. Païdoussis, A.K. Misra, A CFD-based model for the study of the stability of cantilevered coaxial cylindrical shells conveying viscous fluid, Journal of Sound and Vibration 176 (1994) 105–125.
[9] M. Amabili, R. Garziera, Vibrations of circular cylindrical shells with nonuniform constraints, elastic bed and added mass. Part III: steady viscous effects on shells conveying fluid, Journal of Fluids and Structures 16 (2002) 795–809.
[10] A. Lakis, A. Laveau, Non-linear dynamic analysis of anisotropic cylindrical shells containing a flowing fluid, International Journal of Solids and Structures 28 (1991) 1079–1094.
[11] M. Amabili, F. Pellicano, M.P. Païdoussis, Non-linear dynamics and stability of circular cylindrical shells containing flowing fluid. Part I: stability, Journal of Sound and Vibration 225 (1999) 655–699.
[12] M. Amabili, F. Pellicano, M.P. Païdoussis, Non-linear dynamics and stability of circular cylindrical shells containing flowing fluid. Part IV: large-amplitude vibrations with flow, Journal of Sound and vibration 237 (2000) 641–666.
[13] K.N. Karagiozis, M.P. Païdoussis, M. Amabili, A.K. Misra, Nonlinear stability of cylindrical shells subjected to axial flow: theory and experiments, Journal of Sound and Vibration 309 (2008) 637–676.
[14] M. Amabili, K.N. Karagiozis, M.P. Païdoussis, Effect of geometric imperfections on non-linear stability of circular cylindrical shells conveying fluid, International Journal of Non-Linear Mechanics 44 (2009) 276–289.
[15] M. Paak, M.P. Païdoussis, A.K. Misra, Nonlinear dynamics and stability of cantilevered circular cylindrical shells conveying fluid, Journal of Sound and Vibration 332 (2013) 3474–3498.
[16] W. Flügge, Stresses in Shells, Springer-Verlag, Berlin, 1960.
[17] J. Laufer, The structure of turbulence in fully-developed pipe flow, NACA Technical Note 2954, 1953.
[18] F. White, Fluid Mechanics, McGraw-Hill, New York, 1986.
[19] S.E. Haaland, Simple and explicit formulas for the friction factor in turbulent pipe flow, ASME Journal of Fluids Engineering 105 (1983) 89–90.
[20] K. Washizu, Variational Methods in Elasticity and Plasticity, Pergamon Press, New York, 1972.
[21] D.A. Evensen, M.D. Olson, Nonlinear futter of a circular cylindrical shell in supersonic flow, NASA TN D-4265, 1967.
[22] D.A. Evensen, M.D. Olson, Circumferentially travelling wave futter of a circular cylindrical shell, AIAA Journal 6 (1968) 1522–1527.
[23] J.H. Ginsberg, Multi-dimensional non-linear acoustic wave propagation. Part II: the non-linear interaction of an acoustic fluid and plate under harmonic excitation, Journal of Sound and Vibration 40 (1975) 359–379.
[24] P.B. Gonçalves, R.C. Batista, Non-linear vibration analysis of fluid-filled cylindrical shells, Journal of Sound and Vibration 127 (1988) 133–143.
[25] V.B. Nguyen, M.P. Païdoussis, A.K. Misra, A new outflow model for cylindrical shells conveying fluid, Journal of Fluids and Structures 7 (1993) 417–419.
[26] M. Hénon, On the numerical computation of Poincaré maps, Physica D 5 (1982) 412–414.
[27] H. Ziegler, Principles of Structural Stability, Blaisdell Publishing Co., Waltham, MA, 1968.
[28] Wolfram Research, Inc., Mathematica, version 7.0, Wolfram Research, Inc., Champaign, IL, USA, 2008.